



Fixed Income

Fixed Income Framework

NOVEMBER 2024

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1 Introduction

Fixed Income markets play a pivotal role in the global financial system, providing a platform for borrowers to raise funds through the issuance of debt securities with regular (fixed) scheduled payments (hereby referred to as fixed income assets). These markets encompass a wide range of instruments including bonds, notes and commercial paper, that facilitate the flow of capital from investors to entities that require financing.

The introduction of new technologies such as distributed ledger technology (DLT) and asset tokenisation, as well as the ability to represent and process securities digitally are showing transformative potential to enhance the efficiency and access of financial markets. Consequently, financial institutions are increasingly¹ looking at trialling the concept of tokenised fixed income securities, representing conventional debt securities with fixed income, such as government, corporate and municipal bonds in the form of digital tokens.

Tokenisation of fixed income instruments could realise tangible benefits regarding transparency, efficiency in trading and reduction in settlement times across market participants. However, the rapid development in this sector has necessitated the need for development of robust standards to ensure security, interoperability and regulatory compliance at a global level but with respect to local nuances.

Pilots have been conducted over the past decade, exploring the technical feasibility and commercial viability of registering, transferring, settling, safekeeping and trading of tokenised fixed income securities. These pilots demonstrated the expected benefits achievable with asset tokenisation. However, despite its promise, the overall transaction volume and adoption of tokenised fixed income securities in financial services still falls behind expectations. Some key reasons have been identified as follows:

- **Concerns over viability of the business case**
 - High cost of investments needed to start.
 - Lack of Incentive to switch from well-established and relatively efficient infrastructures.
 - Absence of cash on chain to realise lifecycle automation.
- **Regulatory and legal uncertainty**
 - Potentially challenging to fit implementation under existing legal and regulatory frameworks.
 - Different jurisdictions may have different interpretations on the treatment of tokenised fixed income securities that could lead to legal and compliance challenges, particularly for cross-border transactions.
- **Market infrastructure and technology maturity**
 - Significant concerns about scalability and efficiency in processing large transaction volumes.
 - Limited compatibility with traditional market infrastructures implies that significant development effort and investment is needed.
- **Lack of professional custody solutions for tokenised securities**
 - Contrary to what is sometimes claimed, customers expect to hold their tokenised securities through licensed custodians like other traditional securities.

¹ ICMA's tracker on the growing number of new FinTech applications in the bond market.

- Support for corporate actions such as interest payments for a fixed income asset is mostly a manual process, therefore making investment in tokenised securities less attractive.
- **Market readiness**
 - Tokenised fixed income securities, as an emerging market have limited liquidity, when compared to traditional fixed income securities. This could potentially lead to volatility in pricing which is undesirable for the relatively stable traditional fixed income markets.
 - Limited access to a mature / traditional secondary market liquidity through efficient interoperability mechanisms.
 - Heterogenous legal, technical and operational frameworks used by participants.

Project Guardian is set up as a collaborative initiative between policymakers and financial institutions to enhance liquidity and efficiency of financial markets through asset tokenisation; delivered via several different industry groups. This report introduces the **Guardian Fixed Income Framework (GFIF)**, as an initial set of non-prescriptive standards and industry best practices for tokenised fixed income assets to enable the adoption of asset tokenisation for fixed income products within financial services.

This report will provide an overview of the following:

- **Application of tokenisation in the fixed income market**, including differences with traditional models and benefits in each stage of the fixed income lifecycle.
- **Industry standards** for different types of tokenisation and new interaction models allowing commercial scale.
- **Design considerations and best practices for implementation** including data taxonomy, risk disclosures, governance models.
- **Provide practical examples of the implementation of standards** with commercial use cases, guided by policy considerations and frameworks.

The adoption of a common set of standards and industry best practices reduces the incremental investment costs of deploying solutions. The ability to gain critical mass quickly facilitates an expansion of business opportunities and addresses concerns around scale when measured against traditional market practices.

By describing design considerations and best practices, the industry group aims to support the growth and maturation of tokenised fixed income assets, with a view to ultimately enhancing its appeal and accessibility to a broader range of investors, issuers, banks and services providers.

For the purposes of this paper, we will use the following key terms:

- **Digital Twin:** Fixed Income securities that are issued and custodied traditionally, but also converted onto a distributed ledger through digital twin tokens that convey ownership interests in the underlying traditional securities;
- **Digital Native:** Fixed Income securities that are issued and custodied natively on a distributed ledger only (which constitutes the golden source of truth in relation to ownership rights), and therefore do not have traditional securities as an underlying basis;
- Digital Twin and Digital Native instruments are collectively referred to as Tokenised Securities;
- The terms 'DLT' and 'blockchain' are used interchangeably in this paper.

2 Background

2.1 Overview of fixed income market

The fixed income market can be grouped into two main markets, the primary and secondary markets. Examples of fixed income assets that are traded include:

- **Bonds:** Debt securities used by various entities to raise capital. Key types include:
 - Government bonds: Issued by national governments, these are considered low risk investments used to finance government spending and fiscal deficit.
 - Supranational bonds: Issued by organisations like World Bank and International Monetary Fund (IMF) to finance large scale international development projects such as infrastructure development or addressing socioeconomic or environmental issues.
 - Corporate bonds: Issued by corporations to fund business activities typically offering higher yields in exchange for increased risk.
 - Municipal bonds: Issued by local governments or municipalities to finance large capital expenditure, often with tax free interest for investors.
 - Covered bonds: Typically issued by banks and collateralised by a pool of high-quality assets, providing investors protection against issuer risk.
- **Commercial Paper:** Short term secured and unsecured notes issued by corporations and generally used for short term liabilities.
- **Repurchase Agreements, referred to as Repos:** Allow for liquidity management by facilitating short-term borrowing with debt securities being held as collateral.
 - Play a prominent role in structuring of a transaction, flow of capital, liquidity provision and risk management.
 - Within the Debt Capital Market (DCM) space, this is an essential component in mitigating credit risk as the lender has claim on the debt securities that have been bought as part of the repo and can sell the security to recover the borrowed funds.

Key participants within the fixed income market include:

- **Issuers:** The entity issuing the relevant instrument to raise capital
- **Investors:** Individuals or institutional investors such as a pension or mutual fund that will purchase the relevant instrument.
- **Underwriters:** Financial institutions that assist the issuer in structuring, pricing and selling the instrument.
- **Credit Rating Agencies:** Organisations such as Fitch, Moody's and S&P that assess the credit worthiness of issuers and the associated instrument and grade accordingly. Investment Grade Bonds are typically issued by the most credit worthy companies.
- **Market makers:** Firms who provide liquidity in connection with the secondary market trading of fixed income instruments.
- **Custodians:** Financial institutions who are responsible for ensuring the safekeeping and record-keeping of the instrument as part of their servicing.

2.2 Existing industry standards

Alongside the evolution of the fixed income market, associated standards and frameworks have been developed to ensure transparency, efficiency and risk management. They play a crucial role in fostering investor confidence and the smooth function of the fixed income market.

Standards	Examples
Messaging and communication	• SWIFT Messaging (notably for post-trade and asset servicing). • FIX Protocol (notably for trading).
Standard identifiers for instruments and parties	• Examples of instrument identifiers include International Securities Identification Number (ISIN), Committee on Uniform Securities Identification Procedures (CUSIP). • Legal Entity Identifier (LEI).
Settlement and clearing	• Settlement / Messaging across participants (e.g. SWIFT). • Central Counterparty Clearing House (CCP). • Central Securities Depositories (CSD). • Settlement cycles – T+2 within the EU (as of the time of research) and T+1 in the U.S.
Regulatory compliance and reporting obligations	• Regulatory compliance and reporting are essential for market integrity, investor protection, and financial stability, with requirements differing for traditional versus tokenised debt securities. Examples include: - MFID II Pre and Post Trade Reporting. - Financial Industry Regulatory Authority's (FINRA) Trade Reporting and Compliance Engine (TRACE)².
Industry best practices and legal documentation	• Securities Industry and Financial Markets Association's (SIFMA) Master Repurchase Agreement (MRA)³. • International Capital Market Association (ICMA) guidelines for issuance and trading of debt securities, repurchase agreements, as well as sustainable finance including: - Primary Market Handbook (PMH)⁴. - Secondary Market Rules and Recommendations (SMRR)⁵. - Global Master Repurchase Agreement (GMRA) including a Digital Assets Annex. - Green Bond Principles (GBP). - Social Bond Principles (SBP). - Sustainability Bond Guidelines (SBG). - Sustainability-Linked Bond Principles (SLBP).

Table 1: Examples of standards in the fixed income space

² Reference to FINRA's TRACE document.

³ Reference to SIFMA's MRA document.

⁴ Relating to offers of syndicated international bonds in the primary market, to programmes under which such offers may be made and to euro-commercial paper programmes and trades made under them.

⁵ Relating to secondary market practices, including calculating coupon accruals, trading defaulted securities, interest claims for settlement fails, and the process for issuing and executing buy-ins.

3 Problem statements and motivation

In 2023 the amount outstanding in the global DCM increased by 5.9% year-on-year to \$140.7 trillion USD⁶. The potential size of the market makes a compelling case for asset tokenisation. Tokenisation for fixed income assets has the possibility to offer tangible benefits for market participants including issuers, investors, distributors and data providers.

Tokenisation has the potential to widen the reach of these assets to a more diverse body of investors in the form of fractionalised digital tokens, where permitted under applicable law and contract(s) and subject to compliance with any resulting disclosure requirements. With fractionalisation, the minimum ticket size is reduced, allowing more investors to participate in the market. This promotes broader participation from both institutional and retail investors across geographies. This regional flexibility can be clearly observed in the varying minimum tradable bond sizes across different jurisdictions. For instance, in Europe, the minimum tradable size for bonds is generally set at EUR 100,000, while in the US, the threshold can range from USD 1,000 to USD 5,000. In Asia, there is more variance, with markets like Singapore, Hong Kong, India, and Japan offering different minimum sizes depending on the type of bond and local market regulations.

Tokenisation allows for smaller minimum size and more uniform participation in global markets while maintaining compliance with local regulatory requirements. This flexibility enhances accessibility and diversification by allowing investors to spread their investments across multiple regions and asset types. It also reduces intermediation costs by allowing direct peer-to-peer transactions and democratises capital markets access for issuers by reducing the costs and reliance on use of CSDs, making smaller issuances more commercially viable. Moreover, tokenisation optimises transaction efficiency by reducing settlement times from days to minutes using DLT.

The distributed ledger creates a single, authoritative, and trusted data set accessible to all market participants, serving as a single source of truth (or "golden source") for tokenised fixed income securities. This includes key attributes such as ownership, transaction history, and transaction terms, eventually removing the need for reconciliation between different participants and reducing counterparty risk.

Tokenisation also supports the creation of customised financial products with embedded smart contracts. These contracts automate key asset lifecycle by facilitating multi-party workflows, reducing discrepancies and operational overhead for issuers and financial intermediaries. By automating compliance checks, they minimise the need for administrative staff and can potentially lower transaction fees.

Although transaction fees with tokenisation and shared ledgers have been shown to be lower, significant overall cost reductions will only be realised once these systems reach a scale comparable to traditional markets.

Hence, this report aims to outline frameworks for tokenised fixed income assets, referencing existing regulations to define a scalable, global robust governance framework. To fully realise the benefits, it is crucial to establish unified standards across asset classes and networks, along with open, shared ledger

⁶ Global fixed income markets outstanding increased 5.9% Y/Y to \$140.7 trillion in 2023, while global long-term fixed income issuance decreased 0.1% to \$25.2 trillion. <https://www.sifma.org/resources/news/top-10-takeaways-from-sifmas-2024-capital-markets-fact-book/>

infrastructure and interoperable digital asset applications. The review of related standards is covered in other material such as the Global Layer One⁷ whitepaper and is beyond the scope of this report.

3.1 Differences between traditional and tokenised fixed income assets

The operating model of tokenised fixed income asset differs from traditional models. A summary of key differences between traditional and tokenised fixed income assets within a transaction lifecycle is highlighted in Table 2.

Stage	Traditional Model	Tokenised Model
Primary Market		
Issuance	- Issued by a central authority (e.g., banks, governments) - Physical certificates or book-entry systems	- Instant issuance, lower intermediary involvement - Digital representation of assets on DLT networks
Ownership	- Ownership mostly recorded by central depositories or custodians - Lengthy settlement periods (T+1 or longer)	- Immediate ownership transfer on DLT networks - No or reduced intermediaries needed or allowing for hybrid models - Fractional ownership possible
Custody & Safekeeping	- Centralised custodians handle safekeeping - Risk of central points of failure - High custody fees	- Decentralised and transparent on DLT networks - Cryptographic security - New custody models like self-custody, custodial or hybrid set-ups
Secondary Market		
Trading	- Limited to market hours - High broker fees - Settlement delays - High liquidity and global connectivity	24/7 trading & settlement on platforms - Peer-to-peer trades without intermediaries - Fractional liquidity
Settlement	- Settled through clearing houses - Involves multiple intermediaries (clearing agents, brokers)	- Instant settlement (dependent on fiat tokens / instant settlement mechanism) - Smart contracts automatically transfer assets and payments

⁷ The Global Layer One (GL1) initiative published a whitepaper in June 2024 that discusses the role of a shared ledger infrastructure compliant with applicable regulations and governed by common technological standards, principles and practices, on which regulated financial institutions across jurisdictions could deploy tokenised fixed income securities.

Stage	Traditional Model	Tokenised Model
Asset Servicing – Corporate Actions including coupon payments	- Coupons paid periodically (e.g., annually) - Delays in payments	- Automated workflows across participants via smart contracts - Transparent payment schedules on DLT networks - Programable assets enabling modular services
Regulatory & Compliance	- Administrative overhead - Paper-heavy process - Varying but well-established requirements by jurisdiction	- Possibility to enforce compliance rules through smart contracts - Missing unified and clear regulatory standards across jurisdictions
Reporting & Auditing	- Centralised record-keeping - Requires manual auditing and reporting - Complex reconciliation across intermediary chains	- Immutable records - Automated audit trails and real-time reporting - Single source for reconciliation
Maturity and Redemption		
Maturity & Redemption	Redemption processed manually via intermediaries	Immediate redemption at maturity

Table 2: Summary of key differences between traditional and tokenised fixed income assets

3.2 Changes in operating practices and risk considerations

The introduction of tokenisation changes operating practices in fixed income markets. This requires financial institutions to review existing processes, to ensure compliance with relevant regulations and corresponding risks managed. Care should be taken to review the changes to the settlement processes, impact of automation using smart contracts, and feasibility of migrating between on-chain and off-chain.

Changes in settlement processes

Tokenisation reduces counterparty risk and operational costs through the enablement of near instantaneous settlement times logically resulting in no unsettled positions. However, implementation is not without its challenges such as:

- **Settlement netting:** “Near” instant settlement should logically dictate that netting should not occur; however, if desired a useful midway resolution between traditional T+2 and instant settlement could be utilising intraday settlement.
- **Implement a pre-funding solution:** Pre-funding has a significant impact on the reduction of settlement risk by ensuring counterparties have sufficient liquidity prior to settlement. Therefore, a practical implementation that works in conjunction with instant settlement is desirable. Instant settlement requires investors to own cash on ledger (in the case of a buy order) or security tokens (in the case of a sell order) or alternatively have a settlement mechanism bridging instantly into traditional payment systems.

- **Liquidity pools:** It should be noted that liquidity pools may be fragmented (spread across multiple platforms rather than a single unified platform) which in turn can pose significant risks and/or challenges from a pre-funding's perspective for sourcing the right liquidity in a cost effective and timely manner. Several mitigation measures that could be potentially explored include using liquidity aggregators, utilising standardised tokens, or leveraging automated market makers (AMMs) to provide deep liquidity for a wide range of digital tokens.
- **Time Sensitivity:** With the advent of instantaneous settlement, where the reconciliation process identifies a break, there is a strong possibility that the relevant position that breaks has been traded away already. As breaks are picked up, multiple updates will be needed across the network that can have a significant downstream impact.

Automation of key functions

Smart contracts can automate many functions that are traditionally managed manually, such as:

- **Price discovery:** The inclusion of a liquidity pool within the smart contract can include an AMM that determines the price of assets based on the ratio of tokens within the pool. This will fall outside the scope of this report.
- **Issuance:** Issuance procedures can be handled programmatically, reducing time and operational risk.
- **Security and paying agent roles:** Automated lifecycle management and disbursement of payments such as interest or dividends.
- **Tax treatment and voting:** Complex tasks, including tax handling and token holder voting, can be embedded into tokenised processes for more efficient and transparent operations.

Smart contracts make a significant contribution towards the automation of processes such as price discovery, issuance, redemption, and calculation and paying agent functions. Smart contracts similarly can support different taxation scenarios (for example, withholding tax or stamp duty) as well.

Materialisation / Dematerialisation

Whilst tokenisation raises few hurdles for materialisation, the benefits of fractionalising assets can raise challenges for dematerialisation.

- **On-chain vs Off-chain:** As tokenised fixed income securities move between digital and traditional financial structures, the process of bringing assets on-chain and off-chain back to traditional financial systems needs careful handling. For instance, transferring a US\$20,000 position from a digital exchange to a traditional exchange may not be possible if the traditional exchange adheres to minimum ticket sizes of US\$200,000.
- **Scalability of tokenised fixed income securities:** While tokenisation offers higher operational efficiency by lowering transaction costs, increasing liquidity and collateral management for debt instruments, scalability and fractional liquidity remain an issue for broader industry adoption.

By successfully navigating corresponding changes, the industry can reduce operational risks and enable greater efficiency in managing tokenised fixed income securities at scale.

3.3 Frameworks and efforts to establish industry standards

The establishment of product and infrastructure standards, particularly in fragmented global markets, is essential to realising the full potential of tokenised fixed income securities. Existing standards for debt securities should be adapted to ensure seamless integration between traditional and digital

markets. This section outlines current initiatives aimed at developing industry standards and frameworks.

Legal and regulatory frameworks

In recent years, policymakers have adapted existing laws and regulations, or passed new laws to pave the way for tokenisation of debt securities or the use of DLT for lifecycle processing⁸. Examples include:

- **Singapore:** Securities and Futures Act (Singapore).
- **Japan:** Financial Instruments and Exchange Act (Japan).
- **Europe:** Pilot Regime for DLT Market infrastructures.
- **Germany:** Electronic Securities Act (2021) and regulatory guidance from BaFin.
- **Switzerland:** DLT Act (2021) with the introduction of DLT securities and the DLT trading facility as a new financial market infrastructure for the multilateral trading of DLT securities.
- **USA:** SEC guidance on special purpose broker dealers (2020), additionally there are some state-level legislations on tokenised fixed income securities.
- **UK:** The FCA and Bank of England have encouraged firms to trial tokenised fixed income securities in their regulatory sandbox. These assets are classified as investments under the Financial Services and Markets Act, subjecting them to the same regulations as traditional securities. The UK Law Commission is also reviewing existing laws to ensure they are robust enough to handle the unique characteristics of digital assets.
- **France:** Pacte Law, enacted in 2019 provides a framework for the issuance and trading of tokenised fixed income securities. Compliance of these regulations is overseen by the AMF (French Regulator).
- **Luxembourg:** Legislation has been passed that facilitates the use of DLT for transfer of tokenised fixed income securities (2019).
- **India:** Operational guidelines for 'Security and Covenant Monitoring' using DLT (2022), issued by Securities and Exchange Board of India (SEBI).

Fixed income tokenisation standards

Tokenised debt markets can only scale with interoperability across DLTs and traditional markets to prevent liquidity fragmentation and leverage existing standards. To achieve this, industry associations are working to establish standards to promote tokenisation adoption. Examples of industry associations in the fixed income space include:

- **The Capital Markets and Technology Association (CMTA)**⁹, is a Swiss based independent association focused on creating common standards around tokenised fixed income securities using DLT.

⁸ ICMA's DLT Regulatory Directory seeks to provide a non-exhaustive overview of recent DLT regulatory guidance, legislative initiatives, as well as related strategy papers and publications in selected jurisdictions across Europe, North America, and Asia-Pacific. Its aim is to provide a sense of the direction of travel, anticipating future regulatory DLT guidance and legislative change, which will pave the way for broader adoption of DLT.

⁹ In 2022, they published the CMTA Token (CMTAT) an open standard for smart contracts designed specifically for the tokenisation of financial instruments.

- **The International Capital Market Association (ICMA)**¹⁰ have been working on combining existing Bond Data Taxonomy (BDT), published by ICMA in 2023, with new formats and standards suitable for tokenised debt securities.
- **The FINOS Common Domain Model (CDM)**¹¹ standardises the representation of financial instruments through the transaction lifecycle in a machine readable and machine executable format.

¹⁰ ICMA published the Bond Data Taxonomy (BDT), in March 2023, providing an agreed language to promote automation and reduce the risk of fragmentation across bond issuance, trading, settlement and distribution.

¹¹ FINOS, in collaboration with ICMA, ISDA and ISLA, made the Common Domain Model (CDM) open source in February 2023 which covers derivatives, securities lending and repo transactions and by extension bonds.

4 Lifecycle of fixed income securities¹²

The lifecycle of fixed income securities can be divided into three main stages: primary market, secondary market, and maturity and redemption. This chapter of the report provides an overview of how tokenisation may be applied in each stage of the fixed income securities lifecycle.



Figure 1: Lifecycle of a fixed income instrument

In an initial primary market issuance phase, banks, underwriters, and other intermediaries facilitate the issuance of new fixed income securities by governments, corporations, financial institutions and other issuers, and distribute them to investors, be it as part of an issuance programme or within a standalone issuance.

Once distributed in the primary market, these securities are traded among investors and market makers, both OTC and on exchanges, providing liquidity that supports price formation. Securities transactions undergo a clearing and settlement process in the primary and secondary markets, where financial market infrastructure organisations and market participants process trades and legally transfer ownership of securities. During the noteholders' holding period (and thus during both the primary and secondary market stages), custodians ensure the safekeeping and record-keeping of the securities as part of their servicing, which also includes facilitating corporate actions such as the casting of votes in noteholder assemblies and providing compliance and regulatory reporting. A final lifecycle stage consists of managing the maturity and redemption of the securities.

In each stage of the lifecycle of fixed income securities, the implementation of DLT presents benefits and opportunities as well as certain risks. Thus, while tokenised securities could lead to mitigating of operational, counterparty credit and systemic risks, the implementation of DLT should always be operationalised in a manner that meets the high existing standards of regulated capital markets. Safeguards must be ensured for all market participants and mitigants must similarly protect financial markets more broadly.

The following paragraphs set out specific considerations in relation to the implementation of DLT (including tokenised securities) in each of the three main stages of fixed income securities' lifecycle.

4.1 Primary market

The primary market stage of fixed income securities' lifecycle mainly encompasses their issuance and distribution, which marks also the point at which these newly issued securities enter the custody chain.

¹² This section has been prepared by integrating the analysis set out in "Impact of Distributed Ledger Technology in Global Capital Markets" (GFMA, 2023) and "Scaling DLT-Based SSA and Government Bond Markets: A Roadmap Strategy for European Issuers" (AFME, 2024).

Each of these two sub-stages within the primary market are subject to specific considerations when it comes to the impact of DLT adoption.

4.1.1 Issuance and distribution

The primary issuance phase of tokenised fixed income securities is not deemed to be too disruptive, given existing processes are broadly expected to persist. Issuance would, however, act as a necessary on-ramp for the creation of digitally native tokenised securities, driving new workflows to support this activity. Thus, the adoption of tokenisation platforms would support standardisation and elektronification.

Implementation models

In this report, three possible implementation models of DLT-based issuance were considered.

Books and Records (e.g., documents and administration)	Digital Twin	Digital Native
Support pre-issuance workflows, which could provide a starting point for getting institutions and regulators familiar with the technology.	In some jurisdiction, allow greater mobilisation of securities (compared to non-DLT-based formats) as collateral and enable some post-trade efficiency benefits, though these could be limited in primary issuance and are better suited to secondary markets.	Enable the realisation of the broadest benefits case, including significantly lower cost and time to issue, as well as broader security innovation. (in each case in comparison to traditional securities). Smaller and Medium sized corporates and issuers may be able to tap into bond markets more easily, thereby helping to grow capital markets. However, legal and regulatory ambiguity and absence of liquidity are major hurdles.

Table 3: Three implementation models of DLT-based issuance of fixed income securities

Activities and roles

- **Existing issuance workflows are generally highly bespoke and manual.** This adds time and cost to the issuance workflow but is necessary for effective formation of primary markets liquidity.
- **There have been challenges in the standardisation and elektronification of issuance workflows.** This is driven by: mandatory bespoke processes, historical disincentives for transaction manager to support platform intermediation, and challenging behavioural shifts.

Benefits and opportunities

- **The rise of DLT-based asset lifecycle management platforms** could present the most significant market structure change. These platforms are a critical on-ramp to issue Digital Native; adoption may drive standardisation and interoperability, integrating with traditional systems and workflows, and achieving automation benefits.
- **Transaction managers have a low risk of disintermediation.** Their work could evolve to require less operational involvement, but their role in underwriting new issues, distribution and forming

primary markets liquidity will remain. Intermediaries in the primary issuance value chain predominantly focused on operational execution are at a higher risk of disintermediation.

- **Asset classes have highly heterogeneous issuance workflows**, and thus have differing likelihoods of adopting DLT-native issuance. The greatest opportunity exists in transitioning recurring, frequent issuance fixed income securities (commercial paper, medium-term-note programs).
- **Lower cost and time to issue.** DLT-based issuances can be more frequent and faster with an automated issuance workflow. In addition, similarly to any shelf-based issuance such as an MTN program, tokenised securities can be minted at the time of distribution, which is a key benefit to routine or repeated issuances.
- **Customised bond features.** Tokenised securities may hence be easily tailored to investor needs (e.g., frequency and level of coupons etc.) and, therefore, take advantage of market conditions more precisely and effectively. As part of early-stage issuance series, issuers are recommended to deploy smart contracts for key processes such as settlement of primary market transactions and redemption of fixed income securities, in order for them to gain experience with smart contract functionality. Innovative smart contract functionalities can also be preserved in case of fractionalisation (to the extent permitted under applicable law).
- **Green bonds.** The ability to integrate external data, including data related to green or KPI-linked bonds may enrich information dissemination.

Risk management

Despite ongoing development to support the growth of tokenised securities, participation in DLT-based primary markets remains relatively low compared to traditional capital markets, leading to increased liquidity risk, challenging or impairing transaction managers' ability to place initial offerings. However, transaction managers can actively source liquidity and generate interest, alongside broader industry initiatives to pool liquidity in high potential fixed income assets. This can support the formation of DLT-based primary market liquidity and mitigate potential liquidity risk.

As such, DLT is not expected to have a significant role in mitigating existing risk in fixed income primary markets, but it could help mitigate certain operational, legal and compliance risks for market participants.

- **Operational risk:** DLT can mitigate existing operational risk in primary market issuance processes by using smart contracts to (a) coordinate many-party issuance processes, (b) reduce individual errors through independently coded checks, and (c) use multi-party verification processes to ensure individual errors are not propagated. However, these mitigations are likely to be marginal in the near-term, as full impact requires widespread adoption of DLT across multiple participants in the primary issuance workflow.
- **Legal and compliance risk:** Representing the primary market issuance process on a DLT (at minimum the non-competitive portions of the process) can aid transaction managers in ensuring compliance with legal and regulatory reporting requirements and/or improve transparency for would-be investors. The level of mitigation that can be realised is dependent on the degree to which applicable laws and regulations are updated to allow for DLT-based reporting.

Ultimately, the primary market issuance stage of the lifecycle is a key enabler to realise the downstream risk mitigation benefits in the post-trade, and asset servicing and lifecycle management stages of the securities lifecycle.

4.1.2 Custody

The primary market issuance and distribution typically also constitutes the moment when securities enter the custody chain, where they remain until their maturity and redemption. The following considerations in relation to the custody of fixed income securities are therefore relevant to both the primary and secondary market.

Implementation models

The impact of DLT on custody may depend on the implementation model opted by financial institutions.

Books and records only (i.e., for traditional securities)	Digital Custody
• Custody chain participants can use DLT to manage book entries and messages for securities, cash transactions, positions, and accounts, improving reconciliation and account management. • Securities and cash, however, will remain within traditional market infrastructure, with DLT serving as a reference tool. Hence, reconciliation would still be necessary. • In a more developed DLT-based ecosystem, DLT could become the golden source of truth, reducing discrepancies across institutions and reducing back-office costs by standardising data.	• DLT requires safeguarding both Digital Twins and Digital Native tokenised securities in addition to the current account-based custody model. • DLT helps custodians build a platform that meets client rising demand for integrated custody of traditional and tokenised securities, where the beneficial ownership details may be managed by either traditional databases or on DLT- based records. • Custodians' role will evolve to safeguarding private and public keys, which confer the ability to access and transact tokenised securities.

Table 4: Impact of DLT on custody depending on implementation model

Activities and roles

The impact of DLT varies by the type of tokenised securities. The role of the CSD and activities it performs differs according to the nature of the tokenised securities. Hence, this could result in one custody model for Digital Twin and another for Digital Native, reversing the general trend toward consolidation that has characterised CSDs over the past few decades.¹³

- For **Digital Twins**, the CSD's traditional ledger remains as the golden source of truth for the underlying traditional securities holdings, given its legally established nature today. As Digital Twins involve an underlying instrument trading and settling in traditional markets alongside a Digital Twin token on DLT-based markets, the CSD is likely to retain its dual role as the official record keeper of ownership and managing governance issues for traditional securities holdings.

¹³ In ECB's "The securities custody industry" report, they highlighted that developments in the EU may serve to accelerate the pace of consolidation of custody service providers and the trend towards resorting to external providers. Public authorities have also encouraged CSDs to compete in the provision of cross-border services, with the intention of driving down the cost of cross-border settlement as one of the means of achieving a single, integrated capital market.

If internalised settlement occurs, tokens can be transferred on the DLT without updating the traditional CSD ledger. However, the CSD or custodian would likely need to regularly verify that ownership records of the traditional asset match those of the Digital Twin.

- For **Digital Native** assets, the DLT is the golden source of truth. The CSD role may shift toward governance while custodians and other financial intermediaries play a larger role in validating transactions. It should be noted that this model assumes regulation that requires the role of a CSD. Depending on the progression of regulation in different jurisdictions, alternative models may also be possible where a CSD may not be required.

The level of decentralisation is a design decision dependent on several governance and regulatory considerations. First, a consensus mechanism in which custodians or brokers mutually validate their clients' trades could raise competition and disclosure concerns.¹⁴ Custody chain members can potentially mitigate these concerns by implementing privacy-enhancing cryptographic techniques. Second, members need to ensure that the consensus mechanism and associated recourse processes could comply with both legal and governance standards. For example, the CSD role is legally mandated by several jurisdictions (e.g., the United States for certain securities), so any changes to the CSD role resulting from the consensus mechanism would require regulatory blessing.

The financial impact of DLT on custody could be considerable. On the cost side, simplification of reconciliations could lower the total cost to serve. Custody of tokenised securities could open new ways to serve client demand. However, upfront operational and capital expenditure may be necessary to build out custody platforms and (at least in the short/medium term) link legacy and DLT-based platforms.

Benefits and opportunities

- Operational efficiencies and cost-saving impact are realised through the impact of tokenised securities at scale, reducing the complexity of reconciliation efforts. As of 2015, the annual cost base for post-trade processing was \$6–9 billion USD; simplifying these processes could lead to \$2–4 billion USD in annual cost savings, of which reconciliations costs are a major portion.¹⁵ When realised, these savings could be passed up the custody chain.
- Digital books and records could address record-keeping inefficiencies. DLT used across the custody chain could simplify post-trade reconciliations that currently occur between participants, leading to greater operational efficiency through less manual intervention, enhanced data transparency, and auditability. However, DLT may introduce new reconciliations with existing databases as the records become a sub-ledger native to a financial institution's general ledger reporting. This solution would, however, not offer the ability to keep Digital Native in digital Custody.
- Digital Custody could introduce a new infrastructure for DLT-based asset safekeeping via the wallet and key model, relying heavily on private keys for transaction authorisation. To safekeep the private key and establish connectivity with existing accounts, custodians may require new sub-custody or third-party services.

¹⁴ In order to increase transparency and trust in the DLT system, all information on the transactions in the ledgers is typically observed by all system participants and duplicated into their own ledgers. When applied to financial markets, this transparency might cause privacy or competition issues, and thus breach applicable laws, such as the General Data Protection Regulation.

¹⁵ Broadridge believe that by mutualising highly standardised trade processing functions through a utility model —where participants benefit from economies of scale and network effects — banks could wring out up to 40 percent of costs and the industry could save \$2 billion to \$4 billion annually.

- However, any new infrastructure is unlikely to displace the account-based custody model in the near or medium term given the prevalence of traditional assets and presence of tokenised fixed income securities that both require traditional backing, as well as regulatory requirements for security accounts in various jurisdictions.
- Although traditional custody models will remain prevalent, Digital Custody allows traditional custodians to offer an integrated, interoperable custody platform across tokenised and traditional securities. Even if DLT-based market cannibalises current non-DLT market value, they represent growth in new market segment. By investing in tokenisation, custodians can protect their existing assets and capitalise on tokenised fixed income securities growth.

Risk management

Differentiated operational risks could stem from new lifecycle management workflows, such as private key management and maintaining data integrity between distributed and traditional ledgers. Qualified custodians are already implementing mitigations, such as:

- Split keys with quotas for signatures.
- Account abstraction smart contracts, enhancing externally owned accounts like MetaMask.
- Parachute recovery functions sending keys or tokens to a governance address.
- Location-based signing.
- Automated reconciliations with recourse for erroneous transactions.

DLT can further reduce operational risks by simplifying of post-trade reconciliations and minimising manual interventions. However, new challenges in cybersecurity, regulatory compliance and financial crime, business continuity planning, data privacy, asset onboarding, and vendor risk considerations would need to be addressed. Financial institutions will also need to manage risks from running different systems in parallel, adapting workflows, integrating vendors, and ensuring interoperability between distributed and traditional ledger.

- **Operational risk:** The shared ledger can simplify reconciliation effort, reduce positional deviations between organisations and create a standardised data format for post-trade data. A combination of different nodes allows each participant bespoke, programmable access to the shared ledger of ownership. This drives transparency and trust throughout the custody chain.
- **Risk-adjacent impact, operational resiliency:** The distributed nature of the ledger in relation to tokenised securities means greater resilience against operational or system outages, as there is no central operator in certain cases. Existing resiliency and continuity plans have been tested and validated with regulators and can be easily adapted for DLT (e.g., ensuring back-up power/generator and the four-eyes principle). The solid regulatory foundation of traditional, qualified custodians could provide a lower risk means for institutional clients to access tokenised securities, compared with self-custody or other industry options. On the KYC/AML/CFT side, the possibility of custodians accessing client data with the use of digital identity or permissioning will allow drive efficiency at a lower cost.

4.2 Secondary market

The secondary market stage of fixed income securities' lifecycle lasts until their maturity and redemption. The following sections therefore concentrate on the impact of DLT adoption on the trading in fixed income securities as well as their servicing. However, during this stage the securities will also remain in custody, and therefore the considerations set out earlier remain relevant throughout their lifecycle.

4.2.1 Trading

Trading is highly efficient across many asset classes and centralised execution venues, with no immediate disruption to market structures expected. However, as the DLT ecosystem evolves, DLT could enhance infrastructure use throughout the securities lifecycle, including trading. Key innovations like tokenisation could deepen liquidity in traditionally illiquid asset classes and broaden market access. Additionally, new solutions such as automated securities selection, portfolio balancing, and real-time carbon finance tracking could emerge. Despite these opportunities, establishing these new markets may present significant challenges.

Implementation models

Tokenised securities will likely be traded on centralised execution venues and settled on DLT. Decentralised exchanges are broadly viewed as inappropriate for capital markets due to regulatory, technical, and commercial factors. Two types of execution venue may emerge, depending on the type of securities involved.

Traditional fixed income securities	Tokenised securities
Existing workflows can be adapted to incorporate Tokenised Securities into traditional liquidity pools. RFQ-based execution venues can include additional parameters to specify security format, settlement time, and settlement system. Users may have functionality to include the token identifier (rather than just the ISIN) at the point of trade and may have the ability to lock the token on a DLT until settlement.	Specific execution venues primarily transact at a token identifier level, as they are built to natively incorporate features like token programmability, atomic settlement optionality, and fractionalisation.

Table 5: Types of execution venue depending on securities involved

Operating and transacting on a new secondary trading ecosystem would initially incur high costs. To aid market adoption, aggregation of new liquidity pools and integration of workflows into existing systems is critical, with order and execution management systems playing a key role in these developments, enhancing their competitive edge.

While trading activities and workflows will mostly remain unchanged from traditional markets, adaptations are expected. Liquidity is likely to concentrate in select asset classes with clear opportunities and market readiness but will vary across asset classes and tokenisation types.

Activities and roles

The current market structure is unlikely to be disrupted in the near term, but new DLT platforms could emerge and mature as the ecosystem develops. This trading ecosystem is a highly complex array of liquidity pools that vary based on asset class, investor, jurisdiction, and order characteristics.

Existing market structure may be entrenched by the following factors:

- A collapsing distinction between execution venues and clearing and settlement systems, supported in the EU Pilot Regime, through the creation of the role DLT trading and settlement system (TSS), which is an aggregation of DLT multilateral trading facility (MTF) and DLT settlement system (SS) roles. Nevertheless, TSS roles have value in independence as they have distinct market relationships, expertise, and value in the supply chain. The TSS role likely has higher utility in closed trading models and new secondary markets that are not reliant on existing settlement infrastructure or integration with existing CCP netting workflows.

- Compatibility and integration with existing trading infrastructure and independent clearing and settlement providers.
- Workflows and behaviours are difficult to change. New liquidity pools could likely integrate into existing trading workflows and infrastructure.
- Closed trading models and new secondary markets integrated with DLT platforms, operating independently of existing trading workflows and net settlement, may realise consolidation of trading and settlement roles.

Benefits and opportunities

The tokenisation of illiquid asset classes offers the most attractive secondary market growth opportunities. Some examples may include the following:

- **Derivatives:** New features such as automated securities selection for portfolio balancing, interest payment frequency or swaps, ESG tracking, and use of event triggers will enable a faster sourcing of securities.
- **Certainty of settlement:** Optionality for atomic settlement and/or locking tokenised securities on the DLT until settlement could reduce the rate of failed trades, the time delay for settlement and the need to manage risks during legacy settlement periods. This improves trader outcomes and potentially reduces risks and costs.
- **Order routing:** Enhanced use of data allowed by DLT may improve the ability to identify investors and potential sources of the security product in which the buyers seek to invest in.
 - Directed RFQs for corporate bonds to specific counterparties with inventory may reduce information leakage.
 - Liquidity signalling such as masked indications-of-interest may support smart order routing.
- **Increased electronic trading:** Digital security liquidity pools will likely be accessible only electronically, reducing trades executed through voice channels, thus offering potential time and cost savings.
- **Increased liquidity in secondary markets.** DLT enables automated DvP settlement, reducing settlement risk and the time needed for collateral to be locked in repo transactions. This enhances dealers' balance-sheet management by lowering capital costs and boosting secondary-market liquidity.

Risk management

A key risk to address is the potential fragmentation of trading liquidity between traditional and DLT markets, as well as across different DLT platforms. Limited liquidity discourages active trading of tokenised securities, particularly affecting Digital Native securities, which will require new liquidity pools. Secondary market liquidity for these securities may remain poor until critical mass is reached, as the current efforts are focused on DLT infrastructure, primary issuance, settlement, and repos.

Traditional and tokenised securities may also provide an imperfect hedge to support market-making, with a spread likely to emerge as secondary trading activity increases. The potential need to preposition securities and cash for clearing and settlement in tokenised secondary markets could impact liquidity and create a barrier to entry for investors due to pre-funding requirements. Below are several strategies to mitigate these risks.

- Bridging traditional and DLT-based trading venues, and ensuring interoperability between DLT platforms, can resolve market fragmentation and mitigate liquidity risks. On/off ramp solutions

with conversion mechanisms, similar to those used between depositary receipts and ordinary shares, will be key to this process.

- Incentivising liquidity provision and integrating Digital Native securities into existing workflows, such as through the use of AMM to price cross-format risk, can help mitigate risks and enhance participation. Execution venues and market participants should prioritise creating concentrated liquidity pockets, rather than large, low-participation markets. Liquidity risks from pre-positioning can be reduced by setting pre-funding requirements at levels that avoid trapping capital for longer than traditional settlement cycles.

In the long-term, DLT can reduce the number of systems, platforms, and integrations required to access trading markets by providing a unified interface across many asset classes. When combined with existing protocols like the Financial Information eXchange (FIX), this could significantly enhance risk mitigation. However, in the near-to-medium term, progress may be gradual as traditional and DLT-based markets operate in parallel. Nevertheless, as secondary markets for tokenised securities mature, DLT in secondary market trading could enhance risk mitigation across the post-trade lifecycle, in relation to clearing and settlement, custody and asset servicing.

4.2.2 Asset servicing and lifecycle management

Asset servicing and lifecycle management covers corporate actions, tax withholding for interest payments, and regulatory reporting processes, all of which may be impacted using DLT.

Implementation models

As in other stages of fixed income securities' lifecycle, DLT may be implemented using two main models:

Books and records only	Tokenised securities
Records would be maintained on the DLT, while the securities themselves remain on the traditional ledger. The level of impact would be partial compared to a full DLT-based implementation.	This implementation involves the full set of asset servicing and lifecycle management processes for tokenised securities, which necessitates records to be posted and maintained on a DLT.

Table 6: Implementation models for asset servicing and lifecycle management

Activities and roles

Corporate actions: They are legal arrangements that confer certain rights on investors and corresponding obligations on issuers, CSDs, and custodians.

The roles involved in managing corporate actions may evolve in the following ways.

- **Issuers** of securities would remain responsible for initiating corporate actions, but their method of engagement with the markets could change. In addition to the current modes of press release and regulatory filing, issuers could also announce corporate actions on the DLT based platform.
- **Custodians** may see a reduction in operational risk related to corporate action discrepancies and errors in a DLT-based system. However, they remain responsible for overseeing corporate action communications, including monitoring smart contracts that automate these processes. Custodians might also be required to validate new corporate actions submitted by issuers or their issuer agents as part of the consensus algorithm. Local custodians may continue to play an important role in ensuring compliance with local regulations.

DLT could affect corporate action processes in two ways.

- The need for custodians to cross-check multiple data sources for accuracy would be reduced with the use of DLT.
- Smart contracts can automate and shorten processing timelines of corporate actions, assuming requisite integrations are made to clearing and settlement systems.

A smart contract template can serve as an electronic representation of a legal document,¹⁶ with two major impacts on corporate actions.

- Corporate action reference data will be embedded as intrinsic parameters of the tokenised security itself.¹⁷ Corporate actions that are already set out in the prospectus at security issuance can be coded and tokenised immediately.
- The template system forces alignment and clarity on corporate action data.¹⁸ Assuming the template and parameterisation process is based on the legal documentation of the corporate action, the process compels issuer and all other members to explicitly agree to the parameters *ex ante*; there is no room for disparities in interpretation.

If financial institutions implement DLT for corporate action books and records, they can expect impact to be limited to potential processing efficiencies, data visibility and consistency provided by the DLT. However, it would be difficult to realise corporate action execution automation without the use of tokenised securities. Automated execution envisions that corporate action rights and obligations are embedded with the security itself via smart contracts, usually starting at the point of issuance.¹⁹ On a traditional ledger, the process of designing smart contract-based automation for each security would likely be substantially more difficult.

Since coupons are usually set at issuance with minimal customisation, they are ideal for smart contract automation. In this scenario, all stakeholders would connect to a DLT platform where the bond's terms and payment schedules are codified into smart contracts at issuance. As the coupon payment nears, the smart contracts could alert the issuer and paying agent with the terms and calculated payment amount. The issuer then pays the funds directly to the CSD, which distributes payments to investors. This process eliminates the need for reconciliation between the issuer, CSD, and paying agent.

Tax withholding: Core roles are not likely to change. A neutral governing body for DLT-based tax withholding (which could be a consortium consisting of financial institutions and tax authorities) could facilitate the development such a system.

¹⁶ The "Smart Contract Templates: foundations, design landscape and research directions" paper explored a simple semantic framework for smart contracts, covering both operational and non-operational aspects, and describe templates and agreements for legally enforceable smart contracts, based on legal documents.

¹⁷ Digital Asset posted a blog article on "Reimagining the high-stakes, expensive problem of Asset Servicing and Lifecycle Management" which highlighted that the lifecycle data for corporate action can be stored with the asset, visible to participants and updated in real time as a golden-source record.

¹⁸ HSG Task Force highlighted that the adoption of "smart contract templates" could facilitate the expansion of the standardisation of event types and their reference data. These could reconcile the duality between legal and operational aspects of corporate events. By inputting values in a pre-specified template agreed among industry participants regarding a particular corporate event, the issuer would be able to see that information translated into legal prose and verify that it is in line with what its intention.

¹⁹ The key to transforming asset creation and issuance lies in tokenisation and digitisation and the ability to operate across disparate legacy systems and infrastructures, as well as digital exchanges. Smart contracts allow parties to define and agree on a process up front, creating transparent, multi-party workflows. With roles and responsibilities defined, terms embedded in the asset itself, and an accurate record, efficiency is injected into the process.

DLT can streamline tax withholding by accounting for tax treaty benefits and automatically applying the correct tax amount at the source of the taxable event. This reduces the need for reclaim processes and eliminates the reliance on physical documents and wet-ink signatures.

The new DLT-based withholding system could facilitate transparency by providing visibility into investor residency and beneficial ownership, which are two key factors for tax treaty relief. It replaces the linear, sequential Custody chain with DLT, enabling both investors and withholding agents to access and transfer documents quickly. This improved accessibility allows tax authorities to conduct near real-time audits,²⁰ reducing the likelihood of tax fraud.

The extent of impact may vary depending on whether a DLT-based solution can (1) protect confidential data while automating tax treaty eligibility determinations; (2) require changes to a jurisdiction's tax code; and (3) accurately withhold for more complex corporate entity structures; each factor could limit the feasibility of DLT based tax withholding. While the main responsibilities of withholding agents, investors, and tax authorities may not change, the system could require a governing body to align on implementation and promote scalability.

DLT-based tax withholding is a books and records use case. It would require that interest entitlements be represented on the DLT, with personal tax parameters accessible either on or off chain. These records can be entered onto DLT while the securities themselves stay on traditional ledgers.

Regulatory reporting: Regulators and regulated entities likely would not see their roles change in regulatory reporting and could benefit from better data quality and lower operational cost. It remains unclear, however, the impact that constant and automatic vigilance may have on regulated entities.

A DLT-based ecosystem enables embedded supervision, a regulatory conceptual framework that allows supervisor to monitor compliance automatically in real time by reading the market's DLT rather than periodically through large data requests and reports.

While DLT can simplify data aggregation through greater transparency, it does not eliminate the need to comply with data localisation, privacy laws and resolution planning for systemically important institutions.²¹ Financial institutions must still monitor compliance in non-DLT markets and may need to develop integrated systems across traditional and DLT platforms to track their cumulative capital adequacy and prudential positions.

Embedded supervision's impact on broader regulatory reporting could, however, be limited until industry and regulators align on approach to realising three enabling conditions: (1) interoperability among distributed ledgers and with the broader market data, (2) legal guarantee of the integrity of Tokenised Securities, and (3) an established definition of settlement finality, so that the data presented to supervisors is not subject to change.

²⁰ Utilising DLT for administering withholding tax can provide a reliable, confidential and user-friendly way to share information and documentation on a (near) real-time basis across a complex network of intermediaries: registrars, transfer agents, local and global custodians, fund managers, distributors, withholding agents and tax authorities.

²¹ R3 highlighted in their "Applications of Distributed Ledger Technology to Regulatory & Compliance Processes" paper that aggregation will still be necessary. However, the meaning of "aggregation" in this context would change. The process of "combining" data sources from multiple DLT platforms would instead be a process of collecting a series of links that point towards some set of records on a distributed ledger.

Benefits and opportunities

The potential long-term cost savings through the deployment of DLT in asset servicing and lifecycle management are considerable. Corporate action errors cost the custodian industry over \$1 billion USD per year. Unsuccessful tax reclaims cost the industry at least 8 billion EUR per year. Embedded supervision regimes could bring down the cost of compliance, which currently sits at 3–9% of non-interest expense for banks.

The profound degree of change to technology and operations required by DLT-based solutions introduces uncertainty into the investment case.

- In the short-to-medium term, those operating expense savings could likely be incremental, if at all. This is due to the need to run traditional systems in parallel with DLT systems.
- DLT-based solutions may face competition from lower-cost non-DLT alternatives in regulatory technology and process optimisation. These DLT-based systems rely on concerted investments from both public and private stakeholders to achieve network effects. Support from tax authorities and financial supervisors will also be crucial for the success of DLT-based tax withholding and embedded supervision.

Risk management

As DLT is implemented in fixed income securities asset servicing and lifecycle management, new risks could be introduced around data residency, privacy, security, and smart contract execution, given the confidential nature of corporate action, tax, and regulatory data. Permissioning, privacy-enhancing cryptography, and cybersecurity could be paramount to system design. Hence, creating a well-controlled smart contract layer may be crucial to automation in all cases. Market participants may need to establish clear accountability, governance, and recourse for errors in smart contract execution. It should also be noted that DLT platforms can play a crucial role in enabling data consolidation and control.

By using DLT to independently validate and maintain a shared, golden source of truth, data discrepancies are significantly reduced, enabling faster and more efficient corporate actions, tax withholding, and regulatory reporting. Smart contracts further reduce execution risks by minimising manual data manipulation, reconciliation, or data entry. However, these benefits depend on the development of legal and regulatory frameworks that accommodate automated tax, corporate action, and regulatory reporting with use of DLT. Until such frameworks are in place, the full risk mitigation potential of DLT may remain limited.

4.2.3 Clearing and settlement

DLT-based clearing and settlement has the potential to act as an additional, complementary channel alongside existing clearing and settlement infrastructure. This is only expected to impact specific asset classes and transaction types that favour adoption and where secondary market liquidity could pool – not the broader traditional market. For context, clearing and settlement is defined here to include netting, novation (e.g., CCP-related processes), affirmation, confirmation, and allocation (e.g., post-trade processes), and instruction, confirmation, and execution (e.g., settlement processes).

Implementation models

There are four models of DLT implementation within each of clearing and settlement to achieve the execution of DvP, the digital exchange of securities and cash that can enable atomic settlement. The models are derived from the varying role (or lack thereof) that DLT could play across both the securities leg and the payments leg of a transaction.

Implementation model	Description
DLT used as books and records only	No forms of Tokenised Securities nor DLT-based payment instruments. Instead, DLT acts solely as a cross-participant database facilitating updates between participants, but settlement execution and finality remains with the CSD via custodian accounts and must be reconciled on existing systems.
Tokenised securities but no DLT-based payment instruments	Tokenised Securities would settle on DLT, but payment would be coordinated through existing payment systems (e.g., FedWire) or in a commercial bank model (prevalent among international CSDs). This model has already been implemented in the market, integrating DLT platforms with traditional payment systems. For example, there was a World Bank issuance, where Digital Natives called Bond-i were issued but payment was settled using cash in traditional accounts. ²²
Tokenised securities and DLT-based payment instruments settle on different ledgers	Tokenised Securities would settle on one DLT platform, and payment would be coordinated through interoperability with a separate DLT platform. This model would facilitate a form of tokenised commercial bank money (including deposits), and DLT-based payment instruments issued by a central bank (e.g., a CBDC). Market examples of this set-up include Project Stella (European Central Bank and Bank of Japan), ²³ Project Ubin III (Monetary Authority of Singapore), ²⁴ and the Bank of England's synchronisation payments layer concept. ²⁵
Tokenised securities and DLT-based payment instruments settle on the same ledger ²⁶ :	This model could simplify settlement and allow for more flexible use of smart contracts and other DLT-native technologies such as composability.

Table 7: Implementation model for clearing and settlement

Activities and roles

DLT-based settlement could serve as a complementary channel alongside traditional settlement infrastructure, with its impact likely concentrated on specific asset classes and transaction types, such as repos and OTC derivatives. DLT is not intended to drive a market-wide shift to shorter settlement cycles. Most asset classes and transactions will likely continue to use existing clearing and settlement infrastructure, with DLT-based settlement adapting to current cycles and the specific needs of various

²² The World Bank launched bond-i (blockchain operated new debt instrument), the world's first bond to be created, allocated, transferred and managed through its life cycle using DLT. The bond is part of a broader strategic focus of the World Bank to harness the potential of disruptive technologies for development.

²³ Project Stella, a collaborative initiative between the European Central Bank and Bank of Japan, studies the possible use of DLT on financial market infrastructures (FMIs), including large-value central bank real-time gross settlement (RTGS) systems.

²⁴ MAS and Singapore Exchange (SGX) collaborated to develop DvP capabilities for settlement of tokenised fixed income security across different DLT platforms under Project Ubin Phase 3.

²⁵ Bank of England published a background guide on proposed renewed Real-Time Gross Settlement (RTGS) service to deliver a range of new features and capabilities, with Synchronisation being one of those planned capabilities.

²⁶ Adapted from BIS "On the Future of Securities Settlement" and Benos, Garratt, Gurrola-Perez, "The Economics of Distributed Ledger Technology for Securities Settlement".

products, markets, and regulatory constraints. The roles and responsibilities of custodians and market infrastructures will likely evolve with the adoption of DLT in clearing and settlement:

- **Custodians:** As owner and safekeeper of wallets and private keys, custodians could gain significant operational efficiencies in asset classes and transactions moving to DLT-based settlement. Smart contracts can automate the generation, notification, and validation of settlement instructions, streamlining straight-through-processing. Custodians will likely continue handling cash settlement through traditional systems to manage the payment leg. However, if tokenised securities and DLT-based payment settle on the same ledger, custodians may shift to managing conversions between cash and DLT-based payment instruments, provided legal certainty around finality is ensured.
- **CCPs:** In the long-term, CCP processes like netting, could potentially be encoded in smart contracts, enhancing the operational role of a CCP. In DLT-based markets for specific asset classes and transaction types, CCPs could fulfil standard setting and governance functions. Participants could develop distributed financial market infrastructure, where CCP responsibilities for managing default funds and setting margin requirements are shared among market participants via predefined smart contracts or encoded in market-wide infrastructure.²⁷
- **CSDs:** The role of CSD will evolve with the adoption of DLT for specific asset classes and transaction types. In the DLT-based ecosystem, a key open question is around where settlement finality and beneficial ownership will be recorded. If settlement finality is recorded on a DLT, CSDs could evolve to be a governor of DLT-based settlement systems. However, in most models, they are likely to remain a central actor in DLT-based settlement.

Benefits and Opportunities

Given the significant potential impact of DLT on clearing and settlement, its deployment in the fixed income securities lifecycle offer numerous benefits and opportunities in specific markets and/or asset classes. These include:

- **Flexible settlement options:** DLT-based settlement infrastructure could support the industry's long-term shift toward more efficient settlement cycles, helping reduce associated risks. This could lead to a future capital markets ecosystem where settlement speed is tailored to each trade based on liquidity and financial resources. DLT could also facilitate real-time DvP for transaction types like collateral payments or transfers in repos and OTC derivatives. Notably, DLT-based settlement is already being utilised in the intraday repos market today.
- **Synchronisation of clearing and settlement:** DLT could synchronise clearing and settlement into a continuous single workflow, with smart contracts enabling the frontloading of post-trade and pre-settlement processes into a single transaction. This could reduce trade fails in specific markets and/or asset classes where they are a common problem and improve back-office efficiency. Smart contracts would require pre-funding and pre-positioning of securities before trade execution. There are also multiple enablers required including electronification, interoperability with existing systems, and common data standards.
- **Reduced costs:** DLT-based settlement could present operational cost savings and enhance financial resource efficiencies. However, it may introduce new capital constraints due to potential

²⁷ Decentralised Financial Market Infrastructure (dFMIs) proposes a decentralised network of connected nodes representing market participants and collectively responsible for oversight. Market participants interact directly with each other, such that the risk of a transaction is contained within those parties involved in it.

pre-funding requirements and will require significant upfront investment in new technology, especially in the near term.

- **Evolution of CCPs:** CCPs play a critical role in netting transactions and managing counterparty credit risk, which could eventually be automated through smart contracts with proper market conduct safeguards. DLT-based settlement could also enable bilateral and multilateral trading as an alternative to central clearing, offering capital efficiencies and enhanced risk mitigation.
- **Capital efficiency:** DLT-based settlement cycles could unlock capital efficiencies in margin and clearing fund requirements for clearing member firms. The balance sheet capacity generated could create potential for returns on reinvested capital and collateral, estimated in 2016 to be as high as ~\$500M USD in incremental annual revenue in the United States. This amount may be higher today, given the increase in global, trapped, collateral in clearing arrangements.²⁸
- **Reduction of single-point of failure:** DLT-based market infrastructure enables shared responsibility for transaction settlement, minimising single-points-of-failure and concentration risk, unlike centralised systems. The immutable records of ownership transfers also assist in resolving disputes and support the implementation of financial crime requirements.
- **Reduction of settlement failure:** Use of smart-contract execution mandates pre-funding and pre-positioning of securities, enhancing settlement success and reducing costs associated with settlement failure.
- **Reduced reconciliation needs:** Since all processes are conducted on the DLT, it reduces the need for multi-party data reconciliation, reducing the risk of settlement mismatches or errors, both “horizontally” between counterparties and “vertically” through the custody chain.

Risk management

DLT-based settlement may result in higher levels of gross settlement in specific transaction types, potentially requiring more liquidity on hand and increasing overall liquidity requirements at scale, thus contributing to liquidity risk.²⁹ This is compounded by the need for pre-positioning of securities and cash. Legal uncertainties, particularly around the limited alignment across jurisdictions on the legal basis of Digital Native assets and settlement finality poses significant risk, especially in cross-border transactions.³⁰ Achieving DvP on different DLT network types may introduce complex legal challenges that vary by jurisdiction.

In approaching these risks, it should be noted that DLT does not necessitate a gross settlement model and could be configured to support deferred or real-time atomic net settlement. Existing regulatory and risk frameworks can provide applicable guidance for DLT-based clearing and settlement risk management. The use of private-permissioned or public-permissioned networks could also help mitigate risk associated with settlement finality.

²⁸ Assuming a 10% Return on Equity (ROE) (a typical industry target), Goldman Sachs believe the \$5bn in capital savings translates into roughly \$500mn annually in economic savings for broker-dealers.

²⁹ If each tokenised securities ledger had its own cash token, then the need to hold cash tokens on several ledgers could increase aggregate liquidity requirements.

³⁰ The BCBS, in its paper entitled “Prudential Treatment of Cryptoasset Exposures”, has specified that an essential element of classification of a digital asset as a Group 1 asset is that “the applicable legal framework(s) ensure(s) settlement finality” SCO 60.14. As the BCBS states further: “Banks are required to conduct a legal review of the cryptoasset arrangement to ensure this condition is met and make the review available to their supervisors upon request”. Id.

DLT-based clearing and settlement can mitigate a certain number of risks which are present in traditional clearing and settlement, including counterparty credit, operational, liquidity and systemic risks:

- **Counterparty credit risk:** DLT-based settlement can mitigate replacement cost risk due to the use of automation in settlement.³¹ This is driven by the reduction in delay between trade and settlement execution. Principal risk can also be mitigated where DvP occurs atomically, provided the applicable network enables finality as legally defined. If scale is achieved, the notional value of executed but not-yet-settled trades could be reduced, as well as the margin period for CCPs, currently mitigated through capital charges (e.g., collateral). DLT-based settlement could also support participants to post margin in real time. In the case of OTC derivatives, this could address variation margin gaps (between posted margin and calculated current exposure of contractual positions), particularly valuable in times of market stress. Smart contracts could also support the management of margin and clearing funds, with the ledger record of transactions and open positions enabling the allocation of default waterfall more precisely during times of market stress.
- **Liquidity risk:** DLT-based settlement could help mitigate liquidity risk by automating transactions and thus enhance market liquidity, particularly in asset classes like illiquid fixed income.
- **Systemic risk:** CCPs are essential for managing systemic risks and ensuring financial stability. Although CCP failures are rare and effectively mitigated through capitalisation, DLT could complement or offer an alternative model that do not settle through traditional CCPs. For example, using smart contracts and multi-signature accounts, margin, default fund, and additional capital can be pooled at an account controlled by market participants. This ensures that qualified non-defaulting parties can release funds according to a codified waterfall where risk participation is proportional to risk creation.³²

4.3 Maturity and redemption

Tokenised securities can leverage smart contract functionality with an automation of certain workflows and an ability to streamline lifecycle management events. Issuers may therefore deploy smart contracts for key processes of fixed income securities' lifecycle, such as their redemption upon maturity.

The considerations regarding the use of DLT in the repayment of the principal amount at maturity are like those applying regarding the regular coupon payments throughout the lifecycle of the securities, i.e., redemption can occur immediately and in an automated manner via smart contracts.

Smart contracts may also be deployed to initiate the burning of the token representing the tokenised or DLT-native securities, once they have been redeemed. This means that no (or only limited) manual intervention would be required to ensure that the securities are no longer reflected in wallets and/or accounts of investors after reaching their maturity. The automation of maturity and redemption workstreams via smart contracts could drive significant operational efficiency and reduce execution risks within the final stage of fixed income securities' lifecycle.

³¹ DLTs may shorten settlement cycles by streamlining reconciliation processes and reducing the number of intermediaries. A shorter cycle lowers the exposure to replacement cost risk without the need for a CCP.

³² The dFMI is a new concept built on emergent structural truisms of financial markets. This system is socially useful only if incentives are aligned in such a way that risk is assumed proportionally by those who create it and reap its rewards.

5 Operating models

Traditional fixed income instruments are defined by various attributes that allow market participants to identify, categorise, and operate on them. The advent of tokenised fixed income assets introduces new components unique to the digital format.

This chapter identifies design considerations that are applicable for the tokenisation of debt instruments and common archetypes of operating models for tokenised fixed income markets.

5.1 Medium of primary recording

Financial instruments like stocks, bonds, and derivatives have traditionally been issued and traded in physical or paper-based forms, although typically dematerialised into electronic or digital records. The terminology relating to electronic or digitised records of securities should not be confused with “digital assets”, which more specifically refers to assets or rights that are implemented and enabled with the use of DLT. Conversely, the term “Traditional Securities” can be considered a loose reference to historically common forms of securities, both physical and digital, that rely heavily on centralised record keeping and book-entry systems, established financial market intermediaries, and a high degree of manual processing for lifecycle management.

Digital Twins are an evolution of traditional securities, enabling them to cross over into digital assets. They are a form of tokenised security that function as digital representations of traditional securities, issued onto a DLT network. The tokenisation process involves converting ownership and rights of the traditional securities, i.e. underlying real world asset(s), into digital tokens that can be more easily and efficiently traded, settled, transferred, and managed over DLT-based systems. This is a hybrid approach that leverages existing financial products and market infrastructure, combined with advanced capabilities of tokenisation and DLT, to realise several potential benefits such as fractionalisation, instant settlement, peer-to-peer transactions, and greater transparency. This approach does however introduce additional overheads, since Digital Twins must be synchronised with the underlying real-world asset(s). Corporate actions, lifecycle events, data updates, and other changes, to either the traditional underlying or its Digital Twin, need to be appropriately propagated and enacted by the other.

Digital Native on the other hand, is a reference to assets created, issued, and managed entirely on the DLT network, without any underlying real-world asset, or need for traditional market infrastructure to support the asset issuance and lifecycle. Digital Native securities are legally recognised as existing on-chain only, with DLTs serving as official asset registers, and functionality encoded via smart contracts allowing for automated and transparent behaviour. For example, a Digital Native bond might automatically disburse coupon payments directly to registered token holders, in the form of digital payment tokens. Switzerland is one such jurisdiction that recognises ledger-based securities by law, allowing for the direct registration of securities as tokens issued on a DLT register, subject to meeting certain qualifying criteria.

Figure 2 below shows the progressive nature of tokenisation adoption. Starting with traditional securities on new infrastructure, Digital Twins then enabling traditional securities to be migrated onto the DLT network, and finally Digital Native implementations enabling the full potential of asset tokenisation to be realised.

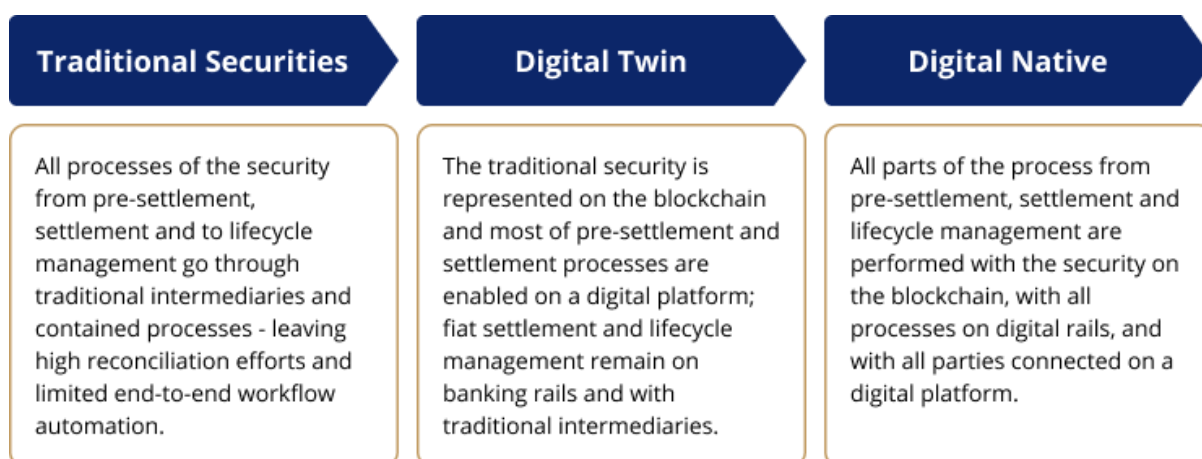


Figure 2: Progressive nature of tokenisation adoption

Tokenisation and securitisation

Tokenisation is characterised by the process through which the ownership of a financial instrument is evidenced by the control (either directly or through a custodian) over a digital token recorded on a DLT. The critical factor is that the ownership of the instrument can be linked to the control over a digital token, like how ownership of a certificated security is tied to possession of the corresponding physical certificate.

In contrast, financial instruments cannot be tokenised if the governing law or the terms of the instrument impose formalities for their transfer, such as a "wet ink" assignment, a notarised document, or registration of the transfer in a centralised registry operated by a single person or authority.

Both fungible securities and non-fungible financial instruments can therefore potentially be tokenised, and tokenised securities can encompass both equity and debt instruments in the broadest sense, i.e. any instruments that incorporate contractual rights or claims, such as structured products, derivatives, interests in contractual collective investment schemes, or structured deposits.

Ownership rights in assets other than financial instruments like co-ownership rights in specific physical assets may also be tokenised, provided that the governing law allows for their transfer via tokens. It is however important to distinguish tokenisation (i.e., the process of associating a right with a digital token recorded on a DLT) from securitisation (i.e., the process of converting property rights in an asset into securities). While tokenisation can complement securitisation by allowing securities issued as part of a securitisation process to be tokenised, tokenisation is no substitute for securitisation as it cannot enable the fractionalisation of property rights by itself.

5.2 Bringing liquidity into new markets

The success of a primary bond issuance and subsequent secondary trading is dependent on the ability to attract liquidity to it. Thus, the ability to bring liquidity into new ecosystems will have a high impact on whether the system will be successful. While the current bond market infrastructure has its limitations, it remains widely used and provides deep liquidity. Over the years, significant capital has been invested in building processes and connectivity. As a result, stakeholders require substantial incentives to migrate to new systems or ecosystems.

A common approach among financial institutions, is to start its implementation with closed-loop private platforms for tokenised securities. This generally requires stakeholders to onboard entirely new ecosystems, abandoning traditional infrastructure. This enables the benefits of tokenisation to be

realised upfront but requires substantial upfront commitment and limits transactions to those coordinated by the institution operating the platform. The closed loop approach has four drawbacks:

- Platform users still need to maintain traditional systems to manage existing assets.
- Existing liquidity in legacy infrastructure cannot migrate over easily.
- Users must set up infrastructure for each institution's individual platform.
- Each institution's private platform would need to interconnect to match the connectivity of traditional market infrastructure.

To address these challenges, financial institutions increasingly create fungibility between digital tokens and assets in traditional infrastructure. Fungibility in this context refers to the ability for a tokenised security to be seamlessly interchangeable with another security of the same type which is issued using a different medium. This intermediation enables ongoing business while providing a pathway to migrate stakeholders to digital platforms. Fungibility requirements need to be addressed on a technical (DLT network and smart contract) and product structuring level. Fungibility will play a key role in transitioning from traditional to digital infrastructure. It allows issuance and trading to function normally while steadily moving stakeholders into the digital ecosystem.

A unified ecosystem broadens the potential investor base by providing investors comfortable with legacy infrastructure an opportunity to participate, and thus enhance liquidity. This is particularly pertinent for hybrid operating models in order to provide participants a robust experience, including (a) flexibility in participating through an existing traditional route or through the digital platform, (b) seamless connectivity between the existing traditional infrastructure and the digital platform without any costs, and (c) transparency and visibility into both security entitlements and trade settlement status.

5.3 Archetypes of tokenised fixed income product deployment

MAS' Project Guardian Open and Interoperable Networks paper examined the design of digital asset networks from four dimensions: (a) Access Layer; (b) Service Layer; (c) Asset Layer; and (d) Platform Layer. It also introduced common archetypes of digital asset networks.

Building upon this foundation, the proposed framework for Tokenised Fixed Income Assets examines common archetypes of potential operating models currently being utilised within the industry. These are summarised in Table 4 below.

Model	Archetypes of tokenised bond issuance	Description
1	Digital CSDs	Traditional CSDs setting up a separate digital securities depository.
2	Extension of Traditional CSDs	Digital assets platforms serve as an extension of the traditional CSD infrastructure
3	Bank Operated Digital Asset Networks	Traditional financial institutions building proprietary tokenisation platforms
4	Multi Party operated Digital Asset Networks	The Issuer or a dedicated institution act as administrator of the smart contract. Digital Asset Network operated by a third party.

Table 8: Illustrative operating model of tokenised fixed income assets

Model 1: Digital CSDs

CSDs seek to continue to support the industry's digital transformation and expand their business model to service investors with access to digital assets.

As regulations and operating models have evolved, CSDs based on DLT registries have been set up as separate entities from traditional CSDs to perform different operational functions. The development of a digital CSD helps to reduce the risk of silos as issuers can choose to perform digital bonds issuances on a single shared ledger platform instead of using different platforms for each transaction. A centralised and standardised system can potentially attract higher liquidity and achieve lower transaction costs as compared to fragmented DLT platforms.

Key Characteristics:

- Lower entry barrier into DLT based assets leveraging existing connections and infrastructure.
- Digital CSDs establish separate depositories specifically for digital securities. These operate in regulated environments and provide high levels of trust for market participants.
- The model focuses on reducing silos and improving interoperability within capital markets, offering a unified platform for the settlement of digital and traditional securities.

Advantages:

- Regulation: Digital CSDs are governed by regulatory bodies, ensuring legal compliance and reducing risks.
- Liquidity & Cost: By centralising the market, CSDs promote access to existing liquidity, reduced transaction costs, and ease of market access for investors.
- Security: With established custody mechanisms, the risk of asset mismanagement is minimised, ensuring investor confidence.

Example:

- A notable application of a digital CSD is the SIX Digital Exchange (SDX) set up by the SIX Financial Group to act as a digital CSD back in 2021. SDX operates under FINMA's supervision, offering a secure venue for the issuance and settlement of tokenised fixed income securities.

Model 2: Extension of Traditional CSDs

Financial Markets Infrastructures are setting up hybrid models to unlock liquidity for digital assets securities and enable a DLT ecosystem. This hybrid approach can take different operating models establishing interoperability between DLT platforms and conventional systems.

The ambition from traditional financial market infrastructure providers and regulators to enable a DLT ecosystem can also be achieved by enabling a hybrid operating model through establishing interoperability between DLT platforms and conventional systems.

Key Characteristics:

- Hybrid models integrate digital platforms with traditional CSD infrastructures, facilitating the coexistence of both traditional and digital assets.
- These models are designed to enable smoother transitions from traditional markets to tokenised fixed income securities by incorporating existing market practices and infrastructure.

Advantages:

- Interoperability: Hybrid models enable interoperability between legacy systems and new digital infrastructures, allowing a gradual shift towards digital assets.
- Scalability: By building on existing infrastructure, hybrid models allow for a cost-effective scale-up in tokenisation efforts.
- Flexibility: Hybrid models cater to both Digital Natives and traditional market participants, enabling flexibility in issuance, settlement, and asset management.

Example:

- DSI (Digital Securities Issuance) is a new service of the D-FMI platform of Euroclear which is a fully regulated DLT platform in production connected to their traditional systems for the secondary market.

The development of a Digital Securities Issuance platform integrated in the existing core services of the CSD helps to accelerate the adoption of digital assets securities by simplifying the onboarding process. It improves marketability and liquidity of the digital security by providing access to trading venues, to collateral services and with the interoperability with other CSDs.

Model 3: Bank-Operated Digital Asset Networks

In recent years, traditional financial institutions have been making significant investments into DLT-based infrastructure. The value of the technology has been proven through several successful experiments in the industry. The focus has now shifted from proof of concepts to commercialisation and development of production-grade systems and infrastructure. As many industry players push into asset tokenisation, there is an uptick in the launch of proprietary asset tokenisation platforms.

Key Characteristics:

- These platforms are developed and operated by individual banks, offering a full end-to-end lifecycle for the issuance, management, and settlement of digital assets.
- Proprietary platforms enable banks to fully control their tokenisation processes, adapting to specific market needs while maintaining compliance with regulatory standards.

Advantages:

- Customisation: Banks can tailor these platforms to specific asset types, issuance procedures, and client requirements, providing a competitive edge.
- Transparency: As these are proprietary, banks have full oversight and control over the processes, ensuring clarity and reduced risks.
- Integrated operations: Proprietary platforms can integrate with other internal systems (e.g., risk management, compliance) for seamless operations.

Example:

- On 29 November 2022, the EIB issued its second EUR-denominated Digital Native bond using Goldman Sachs's GS DAP® as the tokenisation platform. The bonds are governed by Luxembourg law.
- In 2023, HSBC launched HSBC Orion, HSBC's proprietary tokenisation platform for issuance and servicing of digital assets. On 31 January 2023, the EIB issued the market's first-ever GBP tokenised bond, for registration and issuance under Luxembourg, using HSBC Orion.

Regarding such proprietary bank-led/operated new digital asset platforms, there is a high reliance on network effect. Positive developments need to be made to drive liquidity within and among different networks and protocols to attract more participants. As such, there is a need for industry participants to collaborate to increasingly connect existing infrastructures to DLT-based operating models and unlock the liquidity that digital assets bring through investable products bridging both traditional and digital worlds. This set-up also needs to consider how to handle DLT consensus among participants, network governance, participation and other critical network designs considerations. To date most Bank operated networks are operated as closed ecosystems optimising for automation but with interoperability challenges.

Model 4: Third Party Operated Networks

Key Characteristics:

- In this model, smart contracts manage the issuance and settlement of digital securities on open interoperable networks which are managed by a third-party operator.
- Financial institutions or issuers serve as administrators of these smart contracts, ensuring that they meet regulatory requirements while automating various aspects of the issuance and lifecycle management processes.

Advantages:

- Automation: Smart contracts significantly reduce operational costs by automating processes like settlement, transfer, and compliance checks.
- Efficiency: Real-time settlement of transactions, with fewer intermediaries involved, resulting in faster and more efficient operations.
- Flexibility: The ability to deploy on private or public DLT networks allows issuers to tailor the system based on business needs and regulatory compliance.

Smart contract-based issuances are well suited for open interoperable networks as it allows for more flexibility, high degree of interoperability across networks and participants with controls directly implement on smart contract level. Current trade-offs are limited liquidity from traditional markets, standardisation and a sound risk management framework, especially if the smart contract is operated on public networks.

Example:

- A practical industry example is UBS Tokenize which provides a single standardised asset and settlement layer, enabling the origination, management and distribution of tokenised securities. In June 2023, the Bank of China (International) (BOCI) successfully issued a CNH 200 million fully structured dual currency (HKD/CNY) note. This was the first product of its kind in the Asia Pacific region, successfully introducing regulated securities onto a public blockchain, fully constituted under Hong Kong and Swiss law. UBS Tokenize products also showcased interoperability benefits of smart contract-based issuances through the distribution pilot with OSL and the Project Trident.

6 Data model

The approach taken by the **GFIF** is to promote industry standards and notably fixed income protocols and data specifications building on existing standards such as ICMA's BDT. While tokenised fixed income securities embed novel features such as smart contracts to automate coupon payments or (peer-to-peer) transfer of ownership, amongst other processes, each debt security has economic terms at its core irrespective of its form i.e. whether "native digital", tokenised or "traditional".³³

ICMA's BDT, launched in early 2023, has been designed and developed by a broad range of bond market constituents to promote automation and reduce the risk of fragmentation across issuance, trading, settlement and distribution of debt securities. The BDT is technology agnostic and designed to be used both for traditional debt securities as well as DLT-based debt securities.

To this end, BDT provides a common language for key bond information typically contained in a bond term sheet in a (a) standardised and (b) machine-readable (XML) format. The BDT defines unambiguously the data elements of a debt security and DLT platform information that may be communicated between different parties in a transaction (e.g. via APIs, smart contracts, or spreadsheets).

From a technical perspective, each field is defined by its data type, multiplicity and restrictions in the BDT's XML Schema Definitions (XSDs). Many of the fields are defined by ISO standards such as date, currency and legal identifiers. The XSDs' embedded logic includes syntax patterns and predefined enumerations to facilitate accurate and consistent data exchange and straight-through processing, for example, for the announcement of a new issue, allocation and pricing, clearing and settlement, distribution or asset servicing.

The following table provides an example of data specifications in the Bond Data Taxonomy.

- Column A represents data fields typically found in a bond term sheet.
 - Where different terms exist for an identical point (e.g. issuance volume, notional amount, issuance size), the BDT provides a standardised field (e.g. aggregate nominal amount).
 - In the case of DLT-related information where data points are generally not standardised, the BDT provides an umbrella term that enables users to capture relevant information in a structured format (e.g. DLT type, DLT platform operator)
- Column B provides example inputs.
- Column C defines eligible values – in the case of a security identifier, the BDT includes a list of predefined values such as ISIN, DTI, WKN, SEDOL. Currency formats are based on ISO currency codes (ISO 4217).
- Column D defines the data format, for example, string (text) for the issuer name, decimals for amounts, or date formats for settlement date and maturity date.
- Column E defines the technical definition in extensible mark-up language (XML), which facilitates the integration of the BDT into software, data bases, messaging protocols, spreadsheets and other applications.

Importantly, the below examples provide a snapshot and are non-exhaustive. The complete XML specifications and schemas can be found in the Bond Data Taxonomy "pack".

³³ Refer to ICMA's FAQs on DLT and blockchain in bond markets for further information.

Column A: Fields	Column B: Example input	Column C: Field type	Column D: Data type	Column E: XML field definition – xs: type
Issuer Name	Asian Infrastructure Investment Bank; Coöperatieve Rabobank U.A.	free	string	LEIName
Identifier Value	XS2524143554	predefined	string	ISINOct2015Identifier
Aggregate Nominal Amount	300,000,000.00	predefined	decimal	decimal
Specified Currency	EUR	list	string	ISOCurrency (ISO 4217)
Maturity Date	2031-02-22	calendar	dateformat	MaturityDate (ISOdate)
Governing Law	English Law	list	string	GoverningLaw
Gross Up	No gross up if tax deduction or withholding is imposed	free	string	string
DLT platform operator³⁴	Euroclear DFMI, SG Forge	ID-Link	string	string

Table 9: BDT data specification examples

Bridging the gap between traditional debt securities and digital (DLT-based) bonds

While the information embedded in a digital (DLT-based) debt security or stored in a DLT-based infrastructure will vary, the BDT provides a consistent data representation across different platforms and architectures. This facilitates the exchange of data and automation of end-to-end processes from a technical perspective. The [appendix](#) provides an overview of data in the following categories:

- Asset information
- Transaction and settlement data
- Compliance and regulatory data
- DLT-related information
- Contractual terms

6.1 Implementation approaches

This section provides an outline as to how the BDT can be implemented in different token formats, smart contracts and DLT-based platforms, and what data is stored on-chain or off-chain. The Project Guardian Fixed Income industry group's publication aims to develop guidelines that facilitate the use of different protocols and smart contracts based on a consistent data model across the securities lifecycle, both for digital (DLT-based) bonds and traditional debt securities.

Implementation of the BDT can be divided into two approaches as depicted below: (a) integration into software, data bases, systems, or smart contracts whereby the XML structure and schema definitions (XSDs) are used 'natively', and (b) mapping to the BDT's terminology (and structure where applicable) to streamline the exchange of data between multiple parties.

³⁴ Whether a DLT platform is operated by a regulated entity or unregulated entity can be derived from the LEI and further verification.

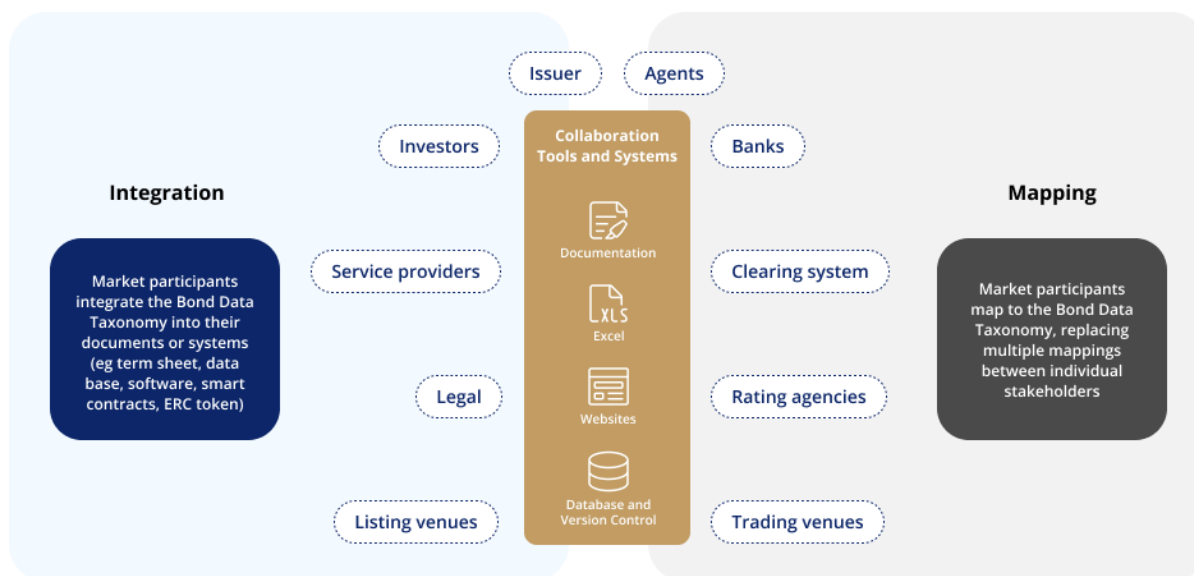


Figure 3: Implementation approach diagram

A user guide including examples as well as a video tutorial are available [here](#).

The following subsection describes how the BDT can be used in a DLT context to define a DLT-based debt security consistently, including different smart contract frameworks and DLT platforms, based on public or private networks. What data is stored on-chain and off-chain, and the use of oracles (external data sources) if required, will be dependent on the use case and notably the type of DLT network and its characteristics i.e. whether private or public, permissioned or permissionless, centralised or decentralised.

6.1.1 CMTA smart contract framework

The CMTA has developed a smart contract framework for the tokenisation of equity and debt securities under Swiss law. However, the standard framework can also be used for private DLT implementations. It follows an Ethereum Request for Comments (ERC)-20 compatible structure with the possibility to add modular add-ons for some instruments and compliance requirements (e.g. allow listing, access rule engine).

Considerations for CMTAT and BDT integration

Under the CMTA draft standard for the tokenisation of debt instruments, a proposal is being developed to reference the BDT in the smart contract framework to define the tokenised debt instrument and to perform certain functions such as coupon payments. This will complement existing functions of the smart contract framework.

The CMTA Token (CMTAT)³⁵ is a digital token standard aimed at simplifying the issuance, trading, and management of securities on DLT platforms. By integrating the BDT from ICMA, industry standards can

³⁵ CMTAT is the product of collaborative work by leading organisations in the Swiss finance and technology ecosystem. It is an open standard for smart contracts to design digital tokens suited specifically for the tokenisation of financial instruments. The CMTAT is a framework that defines both necessary and optional functions that a digital token can implement when tokenising financial instruments such as equity and debt securities, including structured products.

be aligned, making financial transactions faster and more secure. Using BDT terms and XML schema can enhance collaboration and interoperability.

Key insights from ICMA fields show that some are suitable for use with a DLT while others are not. Some fields are managed by issuers, while others benefit from a shared ledger. This setup can lead to improvements like automatic bond ratings on the DLT or blockchain network.

Hence, GFIF suggests a modular contract design approach, like CMTAT's methodology. By using abstract contracts for key field groups, a flexible and scalable system can be created. This allows issuers to structure bonds effectively, leveraging the benefits of DLT while maintaining a robust framework.

For added security, a new document module is recommended to hash bond XMLs and employing a Merkle tree to ensure data integrity. This enhances security and trust for all stakeholders.

In summary, integrating ICMA BDT with CMTAT can improve transparency, efficiency, and security in bond issuance and management, ensuring better compliance and reducing errors through standardisation.

6.1.2 Other BDT implementation examples³⁶

It has been demonstrated that the BDT can be integrated in a modular approach, into various ERC token standards like ERC-20, ERC-1400, and ERC-3643³⁷.

For example, ERC-3643, also known as the T-REX Protocol, is a set of open-source smart contracts facilitating the issuance, management, and transfer of permissioned tokens. It ensures compliance with regulations by verifying counterparties' eligibility during transfers. This token standard expands upon ERC-20 by introducing compliance modules, dynamic identity whitelists, and asset. The framework is compatible with any Ethereum Virtual Machine (EVM)-based DLT, whether deployed on a public or private network.

Both ERC-3643 and BDT aim to optimise digital bond issuance by enhancing transparency and efficiency across the bond's lifecycle. The BDT maps and captures the data relevant to the bond's lifecycle actions, such as issuance, coupon payments, and redemption, into the ERC-3643 smart contracts. These contracts can then apply the specific roles defined in the taxonomy to automate processes, ensuring efficient management throughout the bond's lifecycle from the ISIN generation to redemption.

The bond issuance process starts with the deployment of the Token, Compliance and AssetID smart contracts by the issuer, utilising the BDT on an Onchain Bond Platform. Specific roles are assigned according to the taxonomy, allowing, for example, the Central Securities Depository (CSD) to generate the ISIN code. The system then facilitates order management, enabling the issuance of purchase instructions, matching them with the corresponding securities, and settling transactions on-chain. Lifecycle management follows with automated coupon and redemption payments, which will be handled either by a Paying Agent or through a smart contract, ensuring seamless and transparent bond management.

³⁶ Reference to ICMA's DLT Bonds Working Group webpage.

³⁷ Reference to ERC3643 webpage.

Another example is the proposal by FeverTokens to integrate the BDT into its open-source smart contract package framework and “so|bond v2.0”³⁸. The framework facilitates the implementation of smart contracts under the Ethereum Virtual Machine (EVM) standard, with openly released source code. By embedding the BDT within its architecture, so|bond v2.0 adopts globally recognised standards for bond data, ensuring consistent, machine-readable definitions of bond characteristics such as nominal amounts, interest rates, and maturity dates. This standardisation promotes straight-through processing (STP) and interoperability across different platforms and systems, reducing operational friction and enhancing efficiency.

HSBC Orion is another example of BDT implementation, which incorporates the BDT with a private permissioned DLT platform to represent the tokenised green securities’ key economic terms (e.g., amount, currency, maturity, interest), key dates (e.g., pricing, settlement) and other relevant information (e.g., governing law, parties, ratings).

³⁸ See Sustainable & Open | Bond github at “<https://github.com/so-bond>”.

7 Design principles for tokenised securities

To promote interoperability across different product frameworks and operating models for tokenised debt products, a set of standardised design principles for tokenised securities was developed. These design principles should apply to both Digital Twin and Digital Native Tokenised Securities. They relate to a "smart contract" in so far as it is used to create the token to which a Tokenised Security relates.

The design principles are technology and jurisdiction-agnostic, acknowledging that laws and practices for debt tokenisation in different jurisdictions could vary significantly. Therefore, these design principles must be interpreted, implemented and, where necessary, adapted, in a manner that does not contravene any applicable laws or transaction documentation relating to the issuance of the Tokenised Securities (Transaction Documentation).

The manner in which DLT is used as the underlying technology for Tokenised Securities should not have any impact on the application of these design principles, except that these design principles are only intended to apply where DLT is used to intermediate transfers of Tokenised Securities (or interests in Tokenised Securities) between persons (i.e., they do not apply where an institution's purely internal recordkeeping, accounting, reporting, or other back-office functions are supported by DLT-based infrastructure)³⁹. These design principles use the generic term "Ledger" to refer to ledgers built using DLT, whether such Ledgers are centralised or decentralised and permissioned or permissionless⁴⁰. The term Ledger should be read accordingly.

These design principles aim to promote a common set of standards to facilitate interoperability across different operating models for tokenised debt products, which are further explored in Section 5 above.

Principle 1: Valid existence of the issuer

Recommendation

At the time of issuance and during the life of the Tokenised Securities, each of the issuer of the Tokenised Securities (Issuer) and, if any, the guarantor of the Tokenised Securities (Guarantor), each of which may be a financial institution, a non-financial institution body corporate or a sovereign, supranational or agency, is duly established and validly existing under the law under which it is organised or the treaty or agreement pursuant to which it is organised. Each of the Issuer and the Guarantor (if any) is a separate legal entity that can demonstrate that it has the legal capacity to contract and be sued in its own name.

Principle 2: Validity of the Tokenised Securities

Recommendation

The Tokenised Securities should be duly issued and constitute legal, valid and binding obligations of the Issuer. The Tokenised Securities and their related legal rights should be capable of being attached with, or themselves constitute (if possible, under the governing law of both the Issuer and the Tokenised Securities), a digital token (Token) recorded on a Ledger under the law that governs the

³⁹ For example, these design principles do not apply to situation in which a single custodial institution uses DLT to facilitate transfers of assets among its custodial clients solely on the institution's own books and records.

⁴⁰ To the extent that multiple distributed ledgers are involved, e.g., where one ledger (L2) is built on top of another ledger (L1), the term "Ledger" refers to the top-most distributed ledger used for recording transactions in relation to the Tokenised Securities.

Tokenised Securities, in such a way that the Tokenised Securities and the legal rights relating to the Tokenised Securities cannot be transferred without the Token and vice versa.

Principle 3: Authorisations and consents

Recommendation

To perform their functions in relation to the Tokenised Securities, the Issuer, the Guarantor (if any) and any intermediaries or service providers carrying out activities in relation to the Tokenised Securities should have obtained and will maintain all relevant authorizations and consents, including any licenses, from any relevant supervisory or regulatory authority.

Principle 4: Tokenisation process

Recommendation

The process of tokenisation, including the use of DLT and smart contracts, and the maintenance of any records relating to the Tokenised Securities, should not contravene the terms of the Transaction Documentation and comply with any relevant provisions under (i) the governing law of the Tokenised Securities, (ii) the law under which the Issuer or, where relevant, the Guarantor, is organised or the treaty or agreement pursuant to which it is organised and (iii) the law governing the use of the Ledger in relation to the Tokenised Securities. Any restrictions on the transfer of the Tokenised Securities contemplated by the Transaction Documentation or applicable law (for example, any relevant selling restrictions) must be duly reflected in the functionalities of the smart contract or otherwise be compatible and achievable with the applicable technology platform.

To the extent that the Transaction Documentation does not describe the tokenisation process and its compliance with applicable law, the Issuer should develop or apply a well-defined governance framework for the tokenisation process, in line with the law to which it is subject⁴¹. The governance framework should address (i) the issuance and transfer of the Tokenised Securities, (ii) the choice of the Ledger, and (iii) the creation of the Token and, where relevant, the choice of the smart contract that creates the Token, in each case describing compliance with applicable law.

Implementation

As for traditional debt securities, the Issuer is recommended to have due regard to the principles, standards and best practices developed and recognised by industry bodies, trade associations, or as commonly adopted in the market, which might include recommendations for the Transaction Documentation, including provisions relating to the governing law of the Tokenised Securities and the jurisdiction for disputes in relation thereto.

If the Transaction Documentation restricts the Tokenised Securities' transferability (where permitted under applicable law and technically possible), or if, due to the absence of intermediaries doing the same, the transfer of the Tokenised Securities needs to be monitored by the Issuer to ensure compliance with applicable legal or regulatory requirements (e.g. international sanctions regimes or anti-money laundering regulations), the smart contract or the other technical rules that govern the

⁴¹ These design principles do not address situations in which a party other than the Issuer, such as the custodian or other sponsor of a depository receipts program, is responsible for organising the tokenisation process without the involvement of the Issuer. Additional considerations may be relevant in such situations.

Tokens should allow the enforcement of the relevant transfer restrictions. An example of a transfer restriction would be a restriction on the exercise of any right under the Tokenised Securities until the identity of the Token holder or, if different, of the Tokenised Securities' beneficial owner, has been identified to the satisfaction of the Issuer.

Principle 5: The ledger

Recommendation

In relation to its use for the issuance of the Tokenised Securities, the Ledger should comply with any applicable law and have regard to any applicable principles, standards and best practices developed and recognised by industry bodies, trade associations, or as commonly adopted in the market.

The Ledger should be able to accommodate the following:

- **Power of disposal:** It is recommended that the Issuer accepts in the Transaction Documentation that the holders of the Tokenised Securities have exclusive power, directly or through a broker, custodian or other agent, over the disposal of their legal rights in relation to the Tokenised Securities. The Issuer should permit the delegation of the exercise of such power to third parties such as custodians or proxies. This requirement does not preclude the use of Ledgers where, subject to applicable law and the possibility for technical implementation, (i) the Issuer's consent must be obtained before a Token can be transferred (or where the Token can only be transferred to a purchaser featuring on a "white list" established by the Issuer) or (ii) the Tokens can be transferred without the holder's consent in specific circumstances, for example due to a holder being or becoming subject to sanctions measures. Any such transfer restrictions or forced transfer features should, however, always be in line with the terms and conditions of the Tokenised Securities and any applicable laws governing the Tokenised Securities and the use of the Ledger in relation thereto.
- **Transparency and publicity:** The Ledger should provide, and holders of the Tokenised Securities (or third parties acting on their behalf) should have access to, such data relating to the operation and functioning of the Ledger as envisaged by the Transaction Documents and the terms and conditions (if any) of the Ledger and be consistent with industry practices and applicable law. The Ledger (together with the Transaction Documentation) must also provide adequate information about the Tokenised Securities (and any rights in relation thereto). The terms and conditions of the Tokenised Securities (their nominal amount, maturity date, etc.) must be recognizable from the Ledger (together with the Transaction Documentation) to be tradable. The respective product documentation as well as the functioning of the Ledger must also be clear and transparent. The parties must have access to the relevant entries in the Ledger. This also means that the parties must be able to check the integrity of the entries relating to them without the involvement of third parties (including the Issuer) other than the parties' agents, if any. If entries in relation to the Tokenised Securities are made in more than one Ledger and/or via an off-chain medium, the Transaction Documentation should make clear which record is legally binding on parties (which may be determined by applicable law).

Implementation

The success of this design principle depends on first there being agreement, at the industry level, as to what data should be stored on the Ledger and what data should be stored off-chain.

- **Transferability:** The Tokenised Securities should be registered in accordance with the laws and regulations of the relevant jurisdictions and may be exercised and transferred to others via the Ledger. Finality of transfers taking place via the Ledger should be addressed, taking into account any jurisdictional rules on settlement finality that might apply to the Ledger. The Transaction Documentation should disclose the extent to which settlement finality applies in relation to transfers of the Tokenised Securities on the Ledger.
- **Integrity:** With regard to the integrity of the Ledger, it is required that technical and operational measures are created so that the authorities and powers of each party involved in the operation of the Ledger can be correctly reflected. It would be preferable if unilateral changes to the data and operations of the Ledger were not able to be made; however, in cases where this cannot be achieved, clear governance, processes, and disclosures must be documented. Any person or organised group of persons with administrative authority over the Ledger should operate in an appropriate capacity based on applicable laws, Transaction Documentation and any terms as applicable to the Ledger.
- **Fit for Purpose:** Responsibility and liability for ensuring that the Ledger functions in accordance with and can satisfy its intended purpose, as set out by the Ledger's general terms and conditions (if any) and any other relevant documentation entered into with the relevant participants, should be clearly defined and disclosed to holders of the Tokenised Securities.

Principle 6: Minimum features of the smart contract

Recommendation

The minimum features of the smart contract should address the following requirements:

- **Powers:** Smart contract governance should be defined in terms of possible roles (e.g., Issuer, custodian, tokenisation agent, investor, etc.) with clearly defined entitlements on the smart contracts (which can also be performed through proxies)⁴².
- **Risk mitigation:** Parties should seek to utilise existing technology and operational risk frameworks to mitigate smart contract risk. This can be done through applying existing frameworks to the following eight identified principles to guide best practice in smart contract implementation, each accompanied by practical examples of risk identification and mitigation set out hereafter:
 - Development and growth of internal risk and control and compliance function with appropriate resourcing;
 - Have a clear and proportionate incident response mechanism/policy;
 - Standardised requirements for smart contract audits, including for completeness and robustness while working towards a template-based approach for broader smart contract standardisation;
 - Ensure smart contracts are written in clear, well-documented code that is easy to understand and audit;
 - Extensively test smart contracts before deployment using various scenarios and stress tests;
 - Implement strong access controls to restrict who can modify or interact with the smart contract;

⁴² Where relevant to the Ledger, reference should be made to the Digital Asset Securities Control Principles on this point. The Digital Asset Securities Control Principles were presented in a white paper published by DTCC, Clearstream Securities Services and Euroclear in May 2024.

- Integrate smart contracts with existing traditional operational workflows with human intervention at critical points for added security;
- Agreement of contractual obligations between relevant parties (where applicable).

Implementation

In considering whether a smart contract has the necessary features prescribed under this principle, parties are recommended to have due regard to the principles, standards and best practices developed, recognised by industry bodies, trade associations, or as commonly adopted in the market⁴³.

Principle 7: Tokenisation terms

Recommendation

The Transaction Documentation should take into account the governing law of the Tokenised Securities and its terms should be consistent with the relevant provisions under such governing law, including as to the acquisition, encumbrance, enforceability, or transfer of the Tokenised Securities (in particular the validity of transactions conducted on the Ledger), as well as the validity of the creation or cancellation of the Tokenised Securities and evidencing ownership in relation thereto. It should be clear in the Transaction Documentation the extent to which DLT is used for the creation of the Tokenised Securities and the management of lifecycle events in relation thereto.

To the extent that there are any risks for investors in relation to the use of DLT or smart contracts (having taken into account any mitigating factors), these should be clearly set out in the Transaction Documentation. The rights of the holders of the Tokenised Securities and the way such rights should be exercised should be clear from the Transaction Documentation.

The Transaction Documentation and/or general terms and conditions of the Ledger (if any) should provide for a transparent and fair process for addressing (including cancelling) lost or stolen securities, including because of the loss or theft of a private key. The impact (if any) of the process of the cancellation of a Token on Tokenised Securities should be made clear in the Transaction Documentation or the general terms and conditions of the Ledger (if any). There should also be clearly defined procedures for detecting any breaks in reconciliation and information as to any investigation measures.

Implementation

Due consideration should be given to the tokenisation terms that are developed or adopted by industry bodies, trade associations and other internationally-recognised best practices⁴⁴.

Principle 8: Publication of the terms and conditions of the Tokenised Securities

Recommendation

Subject to compliance with applicable selling restrictions, the Issuer shall make available to each holder of the Tokenised Securities material information regarding the terms and conditions of the Tokenised

⁴³ Such standards include, by way of example, those set out under Annex 1 of the Smart Contract Primer published by the Global Financial Markets Association and the Global Digital Finance available at <https://www.gdf.io/resources/the-smart-contract-primer-an-initial-overview-of-smart-contract-implementation-within-financial-services-regulatory-solutions-for-risk-management>.

⁴⁴ By way of example, CMTA provides tokenisation terms for the tokenisation of securities under Swiss law.

Securities. The Issuer may make this information available through off-chain methods or through use of the Ledger, or a combination of the two.

Implementation

To the extent that an Issuer is seeking to make material information available through use of the Ledger, it may, depending on the Ledger involved, wish to consider the following:

- **Product/Instrument:** Reproducing the terms and conditions of the Tokenised Securities in full in the smart contract, either in plain text or in the form of a "hash" which may be used to authenticate terms stored outside of the Ledger or smart contract. Alternatively, the final version of the terms and conditions of the Tokenised Securities could be published outside of the Ledger or smart contract (e.g. on a webpage or in a third-party repository). Any such reproduction or publication would need to be done in accordance with applicable regulatory requirements that may apply to the Tokenised Securities (e.g. if Tokenised Securities are listed on a trading venue, the rules of that trading venue may require the terms and conditions to be published in a specified manner).
- **Functionality connected with the smart contract:** Recording data pertaining to the key aspects of the Tokenised Securities (e.g. the nominal amount, coupon and maturity date) in the smart contract, should the terms and conditions of the Tokenised Securities not be reproduced in full in the smart contract in plain text or in the form of a "hash". Such data could be included (i) in the smart contract, (ii) in a document stored in an "off chain" repository where the relevant document can be consulted and accessed through the smart contract, or (iii) in such a manner making it possible to identify the relevant documents with certainty (e.g., with information in a DocuSign envelope). It is recommended that the smart contract (and any document accessed through smart contract) should be able to include a clear history and timestamp so any update to it is clearly recognizable. Parties should take into account the principles, standards and best practices developed and/or recognised by the industry bodies, trade associations or as commonly adopted in the market to decide which data should be included.
- **Transparency of terms:** Procuring that the terms and conditions of the Tokenised Securities or other relevant information, where expressed in a machine-readable format, can be translated into terms expressed in a natural language, without any such translation being subject to undue uncertainty or ambiguity

Principle 9: Information on the functioning of the ledger and smart contract

Recommendation

The Issuer should ensure that each holder of the Tokenised Securities can access material information regarding the functioning of the Ledger and the smart contract used for the tokenisation process (which may include the validation process, block ordering and sequencing processes or any other consensus process which eventually results in state updates on the Ledger, and an explanation of the possibility for the freezing or cancellation of the Tokens and the Tokens' key functions) as well as the material technical and organizational measures to protect the functioning and integrity of the Ledger and the smart contract.

Implementation

The material information provided to holders of the Tokenised Securities may, depending on the relevant Ledger and smart contract, include:

- the type of persons eligible to (or the operators of the node(s) that will) validate entries in the Ledger and if the circle of validators is limited by the rules of the Ledger (i.e., whether the Ledger is a public or a private blockchain);
- if the Ledger is a private blockchain, details about how the validators are selected and by whom;
- if the Ledger is a public blockchain, potential risks of 51% attacks or other exploits and procedures to address forks;
- an indication of (i) whether the Issuer may, at its option (including where the Ledger is unavailable for a sustained period of time, for example due to a technical malfunction), request that the Tokens relating to the Tokenised Securities be cancelled and the Tokenised Securities be replaced by a debt security in another form (which may be represented by Tokens recorded in a different Ledger or may be in a form where no Tokens are recorded in relation to the Tokenised Securities) and (ii) where the Issuer has such option, a description of the procedure that would be employed for the cancellation of the Tokens and the replacement of the Tokenised Securities by a debt security in another form (including the procedure to ensure that any information relating to legal ownership of the Tokenised Securities is correctly migrated); and
- an indication that the Tokens are operated through a smart contract and information about the Tokens' properties.

Principle 10: Risk management⁴⁵

Recommendation

Select examples of measures that can be adopted include the following:

- it is advised to have a rigorous governance framework in place for initiation and updating of smart contracts, where used.
- effective control mechanisms should be in place to counter cyber security risks both DLT-specific and more broadly.
- compliance with data privacy and protection requirements for security Ledgers should be in place as well as contingency arrangements to facilitate business continuity.

There should be prominent disclosure in the Transaction Documentation as to any significant risks for investors purchasing the Tokenised Securities, which may include risks relating to the use of smart contracts and DLT. Language as to the mitigation of any such risks may be included to illustrate the level of any such risks, subject to listing and other regulatory requirements and market practice in the jurisdictions in which the Tokenised Securities are being offered⁴⁶.

Implementation

Smart contracts should be extensively tested before deployment using various scenarios and stress tests. Smart contract tests should methodically cover maximum existing use cases and functionalities to minimise unexpected and untested edge cases.

⁴⁵ It is recommended that in relation to risk management and where relevant to the Ledger, the Digital Asset Securities Control Principles should be duly considered.

⁴⁶ See ICMA's publication of "[Considerations for risk factors and disclosure in DLT bond offering documents](#)".

The Ledger and smart contracts should implement strong access controls to ensure only permitted persons can modify or interact with the Ledger/smart contracts. Human intervention should be integrated into the workflows at critical points for added security.

8 Risk considerations and mitigants

8.1 Risk considerations

The ICMA's DLT Bonds Working Group conducted an analysis of offering documents of DLT-based debt securities across a sample of jurisdictions globally in November 2023⁴⁷. In light of the continued increase in the frequency of issuances, the purpose was to foster transparency in this nascent market segment and identify, if possible, any areas of convergence as to the scope and nature of risk factors and other disclosure.

The analysis revealed that issuers have included additional risk factors and disclosures feature in bond offering documents. There are various elements that were found to potentially impact the extent to which the factors are considered for inclusion in an offering document.

The following risk considerations are developed for the purpose of using DLT and smart contracts in a fixed income instrument. As the adoption of tokenisation continues to grow and evolve, these risk considerations are non-exhaustive and will largely depend on the structure of the transaction. These considerations should be suitably adapted so that the relevant risks are appropriately disclosed to the investors.

Area	Sub-Area	Risk Considerations
Technology	Nascent technology and smart contracts	Risks relating to the malfunction, coding error or unexpected functioning of the platform or smart contracts, which may have adverse consequences on the registration, settlement, and transfer of DLT bonds.
		Risks of inherent flaws and limitations, including permissioned networks.
		Risks of technological immutability, such that if processed in error, a transaction to buy or sell a DLT-based debt instrument cannot be reversed.
	Cybersecurity	Risks may increase over time, due to advancements in cryptographic technologies and techniques.
		Risks of manipulation due to malicious actors in the network.
		Risks of exploits or hacks due to failure to update (or update in a timely fashion) a protocol or network.
	Data integrity and consistency	Risks of forking of, failure of, or disruption to, public blockchains.
		Risks of compromised by malfunctioning nodes or errors in the underlying source codes.
	Data privacy and access	Risks relating to accessibility of digital assets such as due to compromise of a wallet owner's private key or due to loss of private key.
		Risks of becoming known to the public which person holds the digital wallet associated with a specific public address.
Legal and regulatory	Jurisdiction variability	Laws and regulations in respect of DLT-based debt instruments being nascent, highly divergent and non-harmonised globally.

⁴⁷ See ICMA's publication of "[Considerations for risk factors and disclosure in DLT bond offering documents](#)"

Area	Sub-Area	Risk Considerations
	Evolving regulations	DLT is subject to a rapidly evolving regulatory landscape (including tax treatment), which might affect the security, privacy, the ability to buy or sell bonds issued using DLT or other regulatory aspects of DLT transactions and trigger changes to, for example, the DLT or blockchain networks and relevant documentation. Future regulatory changes could expose the DLT network and participants to new costs and expenses, which may impact the ability to achieve its business objectives.
Operational	Payment and settlement	Risks related to the use of tokens/representations of fiat currency for payment and settlement, and the conversion thereof to and from fiat currency.
	Interoperability	Risks related to the dependency on allowing interactions between different platforms.
	Third party service providers	Risks related to failures by third party service providers, such as cloud service providers.
Liquidity	Lack of public trust	Risks in relation to the liquidity of DLT-based debt instruments potentially being adversely affected by a lack of public trust in the underlying technology.
	Jurisdiction restrictions	Risks in relation to the inability to list and admit to trading DLT-based debt instruments in certain jurisdictions.
	Existing active market	Risks related to the lack of an existing active market for DLT-based debt instruments
Disclosures	Role of Intermediaries	The role that key intermediaries play (e.g. operator of the DLT platform, custodian, registrar, etc), including in relation to KYC/AML and sanctions checks.
	Type of DLT platform	The types of DLT platform used (e.g. public or private) for the issuance of the DLT-based debt instruments and certain key information in relation to the functioning of such platforms.
	Business continuity	Transparency on business continuity plans.
	Environmental Impact	Environmental footprint concerns of underlying technology.

Table 10: Risk considerations in various areas

Examples of listing and disclosure requirements for Tokenised Securities

Aside from legal and regulatory requirements for issuers to disclose risk factors, fixed income securities that are listed are subject to listing and disclosure requirements by the respective listing venue. A central question is whether a Tokenized Security is considered a transferable security under applicable local laws and regulations. While the answer is subject to structuring choices and applicable securities laws, Tokenized Securities are often subject to specific listing or registration rules, disclosure requirements and restrictions in terms trading on a regulated marketplace. Examples of selected requirements include:

- Confirmation that the Tokenized Security constitutes a security in accordance with applicable law (e.g., governing law under which the instrument is issued, or applicable local law).
- Limitations of pricing and denomination in fiat currency only.

- Information on cash settlement asset (e.g. wholesale CBDC, or other forms).
- Business continuity plans and contingency measures.
- Responsibility and liability statement. Information on risk factors specific to the Tokenized Securities, the underlying DLT network and settlement processes. Information on parties involved, the underlying DLT network, and environmental considerations.

8.2 Risk mitigants

This section outlines various risk mitigations included in the Smart Contract Primer jointly developed by members of Global Financial Markets Association and Global Digital Finance (Primer)⁴⁸. This section should therefore be read together with the Primer.

Risk mitigant 1: Development and growth of internal risk and control or compliance function with appropriate resourcing

Firms need to have effective risk governance structures in place to identify, understand and manage risks associated with the development and use of technology generally, including smart contracts.

An effective risk management system reduces the likelihood of incidents occurring as well as mitigates the impact of incidents.

Such practices also reduce compliance risk—the financial services sector is a highly regulated one, where there are numerous regulatory frameworks/legislation which are technology-neutral (e.g., laws applicable to automation, use of data, artificial intelligence, digital operational resilience and business continuity, may all be applicable to the use of smart contracts, depending on the use case).

The potentially cross-border application of smart contracts also increases compliance risk.

Some examples of existing industry best practice / mitigation strategy:

- Implement and maintain adequate risk management policies and procedures, including procedures for risk assessment, identifying risk tolerances, and risk mitigation.
- Implement and maintain an effective compliance function to monitor, identify, mitigate and address compliance risks.
- Leverage on-chain data to track ownership, using verifiable credentials to enforce compliance while protecting privacy by validating eligibility without exposing private data.
- Ensure risk control and compliance functions should have oversight of, where appropriate, the full development cycle, from proposal to deployment and ongoing monitoring.
- Risk management and compliance functions should monitor compliance of and provide advice to persons/business lines developing, managing, and deploying smart contracts.
- Appoint a chief risk officer and compliance officer and establish relevant committees (with individuals assigned to key roles and responsibilities, and relevant reporting lines).
- Monitor on a regular basis the adequacy of such measures and take steps to address deficiencies.
- Risk control and compliance functions should be directly accountable to the management body.
- Risk control and compliance functions should have the necessary authority, resources, expertise and access to all relevant information.

⁴⁸ Reference to “The Smart Contract Primer: An Initial Overview of Smart Contract Implementation within Financial Services & Regulatory Solutions for Risk Management” report published by GDF and GFMA.

- Risk control and compliance should be independent of the business they control.

Risk mitigant 2: Have a clear and proportionate incident response mechanism/policy

Regardless of what technology is being used (e.g., smart contracts or otherwise), financial institutions may suffer ICT incidents—in the case of smart contracts this could include errors / mistakes in code, software bugs, malfunctioning of related systems or devices, and cyberattacks.

Some examples of existing industry best practice / mitigation strategy:

- Have clear and proportionate incident response mechanism/policies and procedures, including: procedures to identify, track, log, categorise and classify incidents (e.g., by prioritizing based on business criticality); procedures to identify, analyze and solve the root cause behind one or more incidents; processes to mitigate impacts and harm of incidents and to ensure continuation of critical business functions and processes when disruptions occur (including, if appropriate, the ability to continue operating certain functions and services without the use of smart contracts); and post-incident review processes to analyze causes of disruption and need for improvements.
- Maintain up-to-date inventory of smart contracts and relevant ICT systems, devices, databases in relation to which such smart contracts are used, and document interdependencies between different ICT assets to facilitate efficient response to security and operational incidents.
- Integrate smart contracts with existing workflows (see also risk mitigant 7 below) with human intervention at critical points for added security.
- Define, implement, and regularly test data and systems backup and restoration procedures to ensure that they can be recovered as required.
- Establish effective communication plans, both internally (e.g., escalation procedures) and externally (e.g., notifications to stakeholders).
- Design at the outset smart contracts which can be upgraded as the need arises to mitigate any new and evolving risks or to solve post-deployment bugs etc.

Risk mitigant 3: Standardised requirements for smart contract audit

Generally, a financial entity's ICT systems and processes would be audited on a periodic basis, particularly those that support critical business functions and operations.

Given that smart contracts are designed to execute autonomously and may be used in combination with sensitive data, smart contract code should be audited for security issues or errors.

Smart contracts may also consume off-chain data through external systems such as data oracles, which themselves may be susceptible to manipulation—appropriate auditing practices will therefore help to mitigate security and data accuracy risks in relation to such oracles.

Some examples of existing industry best practice / mitigation strategy:

- Ensure governance processes include audits and necessary mitigation steps to address any material or critical risks.
- Ensure audits are undertaken by independent auditors with sufficient knowledge, skills and expertise in smart contracts and DLT.
- Given the transparent nature of DLT-networks (including permissioned networks), consider conducting audits on an automated and constant basis, rather than on a periodic basis.

- Where smart contract code is developed/provided/managed by third party providers, grant appropriate audit and access rights to financial entities and their regulators in the relevant vendor contracts.
- Integrate international standards applicable to software more broadly, such as ISO/IEC 27001.
- Conduct audits in line with smart contract-specific standards which may be developed by industry.
- In addition to audits around software code, carry out audits relating to external data which feeds into the smart contract such as via data oracles, as well as audits on internal governance, risk controls and compliance functions around smart contract development and deployment processes.

Risk mitigant 4: Ensure smart contracts are written in clear, well-documented code

As many smart contracts have self-execution coded in, this could lead to self-execution of errors. This may cause unintended consequences, including the violation of contractual terms and provisions.

Clear and readable code helps to reduce risks of errors in code development, as well as to mitigate risks around code interpretation in the event of a dispute in the context of the associated legal agreement.

More broadly, clearly written and readable code also facilitates code audits, maintenance and updates, which can help to mitigate compliance risks as well as technical/operational risks of failures.

Some examples of existing industry best practice / mitigation strategy:

- Use well-tested programming language and libraries, and established market standards (such as ERC).
- Use a contemporary compiler version (that is, the compiler version is the one assessed in the most recent audit).
- Use consistent coding style such as naming conventions, indentations and commenting.
- Use in-line comments to explain the purpose of a particular line of code and the relevant logic.
- Use meaningful variable names, function names, and clear logic structures where possible to ensure readability of the code.
- Avoid excessively deeply nested structures.
- Integrate smart contracts with existing workflows with human intervention at critical points for added security.

Risk mitigant 5: Extensively test smart contracts before deployment

Lack of pre-deployment consensus of the business logic represented in the smart contracts could result in errant or inadvertent business logic execution.

Erroneous code could result in potential breach of contractual terms, and risk of voidability of contract.

Some examples of existing industry best practice / mitigation strategy:

- Ensure all code is reviewed, tested, and audited as appropriate by third parties.
- Consider “best practice” standards for the review of smart contracts pre-deployment.
- Ensure that smart contracts are written in clear, easy-to-audit, well-documented code (see also risk mitigants 3 and 4 above).

- Extensively test smart contracts before deployment using various scenarios and stress tests with external third parties where appropriate (see also risk mitigants 3 and 4 above).
- Ensure technology and cybersecurity policies allow for code review and testing, where appropriate.

Risk mitigant 6: Implement strong access controls

Lack of smart contract security audit and review process, as well as lack of appropriate lifecycle development programs could lead to potential for exploitation of vulnerable code and flawed business logic leading to denial-of-service attacks, fraud or theft.

Some examples of existing industry best practice / mitigation strategy:

- Implement a standardised secure software development lifecycle that includes smart contract security audit and formal verification of sensitive functions.
- Regularly perform smart contract-focused threat modelling to capture, communicate, and remediate the risk of each use case.

Risk mitigant 7: Integrate smart contracts with existing workflows with human intervention at critical points for added security

Malicious threat actors (with an increased risk of this on public-permissionless networks) could exploit vulnerable code via third-party hacking, leading to fraud/theft.

Some examples of existing industry best practice / mitigation strategy:

- Implement ledger security monitoring and alerting capabilities.
- Leverage a key management solution with transaction level policy enforcement capability.
- Implement a defense-in-depth strategy with multilayer access controls to restrict who can manage or interact with the smart access controls to restrict who can modify or interact with the smart contract and its middleware.
- Implement a mechanism that requires certain access privileges for approving orders that exceed pre-determined volume and price thresholds that are pre-set within the code.
- Implement a procedure for reconciliation between on-chain and off-chain records to detect any erroneous transactions and/or fraud or theft.

Risk mitigant 8: Agreement of contractual obligations between relevant parties

Agreeing to a favorable governing law and jurisdictional provisions mitigates enforceability risk and, in the event of a disagreement between counterparties to an agreement, facilitates quicker dispute resolution by ensuring the jurisdiction of the legal agreement is not in question.

Some examples of existing industry best practice / mitigation strategy:

- Consider the governing contract law principles prior to reaching contractual agreement on governing law and jurisdiction provisions.
- Ensure that all parties have identified the valid smart contract.
- Include contractual provisions allowing for the creation of valid smart contracts that satisfy requirements for legal agreements of that type.

9 Case studies and examples

In this chapter, pioneering case studies, demonstrating the practical application of DLT in DCM are used as references to consider the concepts discussed under GFIF.

9.1 Project Trident

This pilot transaction involved UBS AG (Switzerland), DBS Bank (Singapore) and SBI Group (Japan) collaborating to issue a digital bond with a repo facility on an interoperable network. The aim of the pilot was to evaluate the benefits of DLT for cross-border distribution, financing, and on-chain settlement of capital market instruments. Project Trident provides a revolutionary solution to the issuance, trading, and settlement of financial instruments, setting new industry standards and paving the way for broader adoption in global financial markets. The key roles and responsibilities are:

- UBS: Digital Bond Issuer
- DBS: Order Manager, Bond Investor and Repo Borrower
- SBI: Repo Lender, Stablecoin Issuer
- Public Ethereum: Network Operator

Trident involved the execution of a cross-border repo transaction utilising a tokenised bond with the following key components:

- Three regulated financial institutions in three different jurisdictions (Singapore, Switzerland & Japan).
- Fully automated lifecycle even on-chain (based on smart contracts)
- Legally compliant and fully controlled through whitelisting mechanisms on-chain
- Regulated payment tokens – CHF tokenised deposit and Yen stablecoin
- Utilises public Ethereum as a common interoperable infrastructure

9.1.1 Business process

Issuance of the tokenised bond

- **Issuer (UBS):** The process begins with UBS issuing a digital bond on the Ethereum blockchain. This bond is tokenised, meaning it is represented as a digital asset on the blockchain ledger, with all relevant legal and contractual information embedded within a smart contract.
- **Smart Contract Deployment:** The smart contract is designed to automate the issuance process, ensuring compliance with all regulatory requirements and the automatic execution of contract terms without the need for intermediaries.

Primary market transaction

- **Participant (DBS):** DBS enters the market as a buyer of the tokenised bond. The transaction is conducted entirely on-chain, meaning all aspects of the transaction, including payment and settlement, occur within the blockchain environment.
- **Stablecoin Integration (SBI-JPY):** To facilitate this transaction, SBI issues a stablecoin pegged to the Japanese Yen (SBI-JPY). DBS purchases the bond using SBI-JPY, ensuring a seamless, real-time transfer of value with reduced settlement risk.
- **Smart Contract Execution:** The smart contract governing the bond automatically transfers ownership to DBS upon receipt of the stablecoin, updating the blockchain ledger instantaneously.

Repo transaction process

- **Collateralisation:** DBS can leverage the tokenised bond as collateral in a repurchase (repo) agreement. In this scenario, DBS enters into a repo contract with SBI, using the bond to secure a loan of stablecoins (SBI-JPY). The smart contract automates the terms of the repo, including interest rates and repayment schedules.
- **Blockchain-Based Repo:** The entire repo transaction, from collateralisation to settlement, is managed on the blockchain ledger, providing full transparency and reducing counterparty risk. The use of a tokenised bond allows for instant settlement, enhancing liquidity in the market.

Redemption and settlement

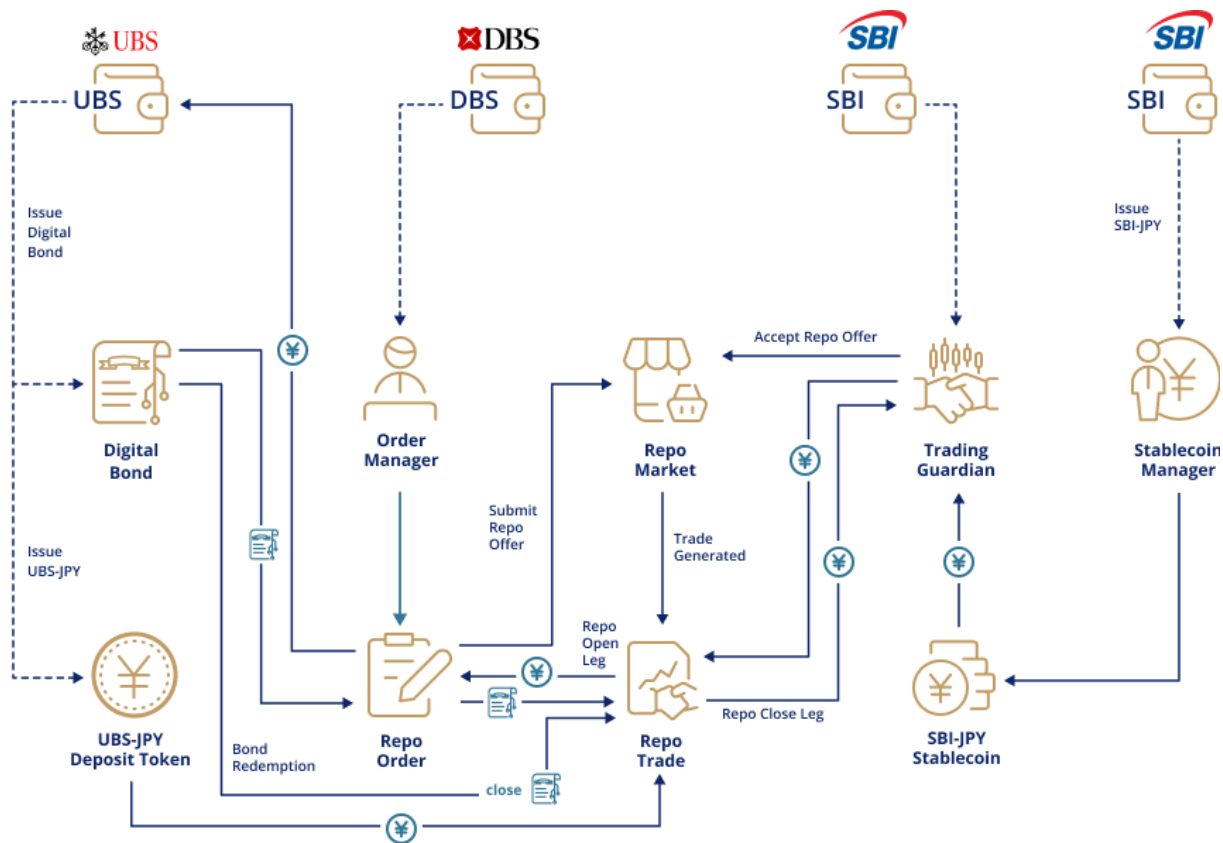
- **Maturity and Redemption:** Upon the bond reaching maturity, UBS (as the issuer) initiates the redemption process. The smart contract automatically executes the terms of redemption.
- **Stablecoin Flow:** SBI, acting as the stablecoin issuer and custodian, facilitates the conversion of SBI-JPY into fiat currency (JPY) for DBS, completing the redemption process seamlessly.
- **Blockchain Integrity:** The use of blockchain technology ensures that the entire redemption and settlement process is transparent, secure, and immutable, with a clear audit trail.

User journey

- **Investor Interaction:** From the perspective of an investor, the process of purchasing, trading, and redeeming a bond has been streamlined through tokenisation. Investors can interact directly with the blockchain ledger, reducing reliance on intermediaries and enjoying faster settlement times.
- **Custodial Services:** SBI provides custodial services for the tokenised bond and stablecoin, ensuring that all assets are securely managed and that transactions are compliant with regulatory standards.

9.1.2 Solution design

- **Tech Stack:** The solution is built on the Ethereum blockchain, utilising smart contracts for the execution of transactions. UBS utilised the CMTAT for the digital bond.
- **Smart Contracts:** ERC standards are employed for tokenisation and transaction execution, with rigorous external auditing to ensure compliance and security.



9.1.3 Key achievements, learnings and future developments

- **CMTA smart contracts:** Project Trident utilises CMTA smart contracts, which adhere to Swiss law, enabling the tokenisation of both equity and debt securities. These smart contracts ensure compliance with regulatory standards and provide a legally binding framework for the transactions.
- **Cross-border compliance:** Seamless execution of transactions across Singapore, Switzerland, and Japan, showcasing robust regulatory compliance in multiple jurisdictions.
- **Global market access:** The integration of stablecoins and tokenised bonds open new avenues for global market access, allowing participants from different regions to engage in seamless, cross-border transactions.
- **Scalability:** The success of Project Trident could lead to the broader adoption of tokenised bonds in the global DCM, enhancing liquidity, transparency, and efficiency across the market.
- **Innovation catalyst:** By proving the viability of blockchain technology in financial markets, Project Trident serves as a catalyst for further innovation in Tokenised Securities and related services.
- **Instant settlement:** By enabling real-time settlement through blockchain technology, Project Trident significantly reduces the time and costs associated with traditional bond transactions, which often take days to settle.

Takeaways	Complexity for Real-World Application & Mass Adoption	How This Could Be Addressed
Smart contracts audited and informally organised across three project teams	Issues of governance and preventing abuse were noted. Full auditing needed across all participants.	Stipulate a clear governance framework for smart contracts representing assets and performing standards. Leverage existing standardisation frameworks like the CMTA, with independent audits and validation of standards.
Payment flows were represented by non-fungible tokens	Exposure of banks to tokens, scaling issues with multiple currencies.	Leverage a mutually trusted and accepted payment token for scalability and reduce pre-funding requirements.
Technical setup only works for exact participants prepared in advance	No scalability beyond pre-defined boundaries.	Ensure interoperability with traditional financial market infrastructures and allow for a level playing field and interoperable technology. Leverage existing industry standards and open-source technology allowing more scalability.
Verifiable Credentials (VCs) used for KYC	Complex and non-scalable, operationally cumbersome.	Solution: Digital identities managed on-chain, with VCs issued by trusted issuers or validators.

Table 11: Observations

9.2 HSBC digital asset pilots

9.2.1 Pilot overview

In collaboration with the Monetary Authority of Singapore (MAS) and multiple stakeholders, HSBC has led efforts to tokenise financial instruments through Project Guardian. The digitalisation of the entire product lifecycle – from issuance to settlement – demonstrates how tokenisation can enhance transparency, security, and operational efficiency. HSBC’s participation in Project Guardian and other projects also highlights the potential for blockchain technology to improve the issuance, trading, and settlement of bonds and other financial products, creating a benchmark for future digital offerings.

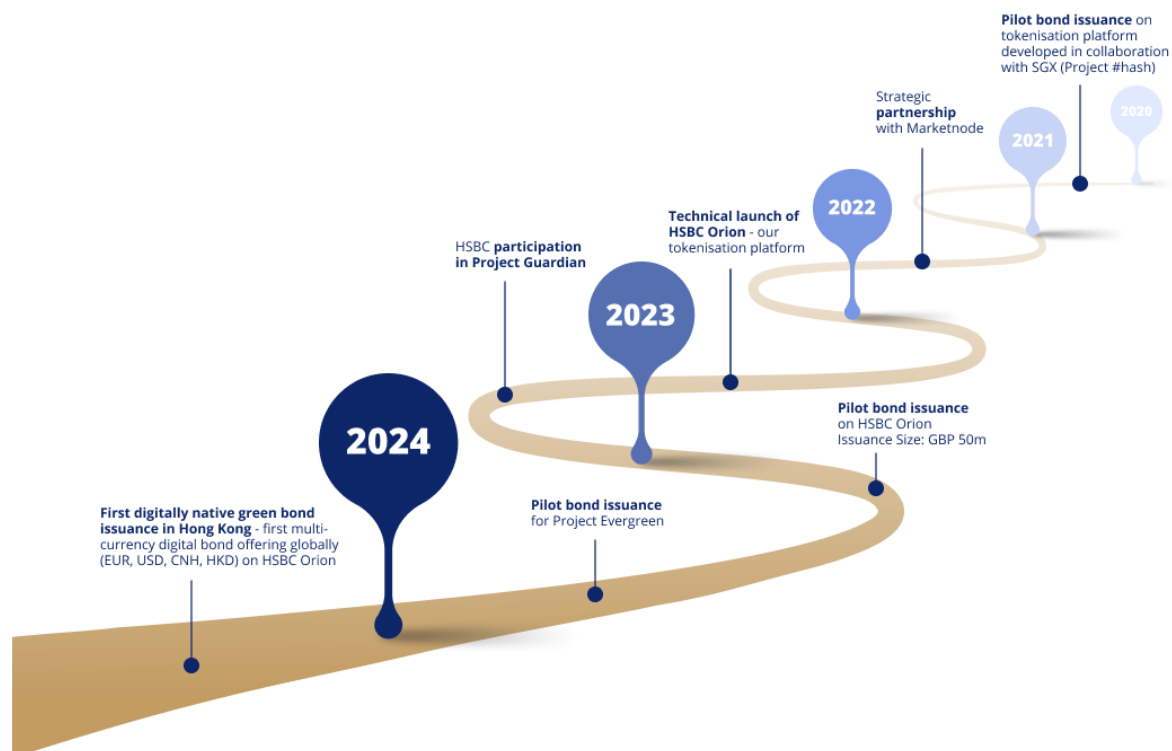


Figure 5: HSBC digital asset tokenisation timeline

9.2.2 Objectives

- **Tokenisation of bonds:** HSBC aimed to demonstrate how tokenised bonds can streamline the issuance and lifecycle management process while maintaining regulatory compliance.
- **Promoting sustainable finance:** Incorporating Environmental, Social, and Governance (ESG) principles into the tokenised bond issuance process to meet rising demand for sustainability-driven investments.
- **Multi-location distribution:** Showcase cross-border collaboration and issuance of bonds denominated in multiple currencies, providing a scalable solution for the global market.

9.2.3 Solution design

- **Proprietary platform (HSBC Orion):** HSBC developed HSBC Orion, a blockchain-based platform designed to facilitate the issuance, registration, and lifecycle management of tokenised bonds. The platform enables the digital settlement and clearing of bond transactions in compliance with local regulations across jurisdictions.

- **Partnership with Marketnode and UOB:** Through collaboration with Marketnode and UOB, HSBC's tokenised solutions aim to expand wealth management offerings, digitising the wealth industry to minimise operational risks.
- **Multi-currency tokenisation:** Tokenised bonds were issued in various currencies (GBP, HKD, RMB, USD, EUR), allowing investors from different markets to participate seamlessly. The use of blockchain technology for these transactions enhanced transparency and significantly reduced operational risks.

Issuer	Government of Hong Kong SAR	Government of Hong Kong SAR	European Investment Bank	Olam International
Pricing Date	Feb 2024	Feb 2023	Jan 2023	Sep 2020
Issue Details	HKD6b digital green bonds	HKD800m tokenised green bond	GBP50m senior unsecured bond	SGD400m senior unsecured bond
Location of the Platform	Hong Kong	Hong Kong	Luxembourg	Singapore
Form of Note	Digital native form. Parties which have direct access to the HSBC Orion can hold the legal titles of the Digital Green Bonds; parties who are not Direct Participants will hold beneficial interests in the Digital Green Bonds represented in their CMU accounts, Euroclear accounts, Clearstream accounts or custodian arrangements, as applicable	Beneficial interest in the Notes was recorded on GS DAP®. The tokenised green bonds were cleared and settled through the CMU.	Dematerialised from and will be registered in the Digital Platform under the Luxembourg legal framework	Similar to conventional notes
Governing Law	Hong Kong law	Hong Kong law	Luxembourg law	Singapore law
Settlement	On-chain settlement	On-chain settlement of beneficial interest	On-chain settlement	Parallel settlement proof of concept
Settlement Time	T+1	T+1	T+2	T+2
Cross-chain Connectivity	No – private only	No – private only	Yes – private only	No – private only

Figure 6: Comparison of Tokenised Securities offerings across multiple issuers

9.2.4 Key achievements, learnings and future developments

- **Scalability of tokenised bonds:** The success of HSBC's tokenised bond issuance underlines the scalability of blockchain technology across multiple jurisdictions and currencies.
- **ESG integration:** The tokenised bond issuance demonstrated the feasibility of embedding ESG principles into financial products, providing investors with verifiable data on sustainability impact and aligning technological innovation with environmental goals.
- **Cross-market collaboration:** The success of the offerings proved the benefits of working across borders and institutions, such as with the Hong Kong Monetary Authority (HKMA), European Investment Bank (EIB), and Singapore's MAS, leading to the first digitally native bond in Hong Kong.
- **Efficiency gains:** The digitisation of the bond lifecycle reduced operational inefficiencies and transaction costs while maintaining regulatory compliance. This also resulted in faster settlement times and greater transparency for investors.
- **Future potential:** HSBC's platform can be expanded to accommodate other financial instruments, such as catastrophe bonds and certificates of deposit as well as funds. The technology has proven to be flexible, reducing costs and time associated with post-trade processes.

9.3 SGX Digital Bond Listing and Digital Assets Register

9.3.1 Pilot overview

SGX Group “(SGX)” has launched the Digital Bond Listing framework and the Digital Assets Register pilot, to help issuers and investors by increasing visibility and transparency of digital assets, with SGX also serving as a “trust anchor” to enhance market confidence.

With the Digital Bond Listing framework, SGX promotes the growth of tokenised traditional financial instruments by including this segment in our existing debt securities listing platform and processes. Issuers of digital bonds will be able to enjoy the benefits of listing, similar to traditional bonds, including greater visibility and investor reach, while investors benefit from greater transparency and assurance that regulatory disclosure requirements are met.

In addition, SGX has launched a Digital Assets Register pilot as a proof-of-concept for security tokens representing debentures. The Register serves as a centralised repository of digital asset issuances, providing a platform where investors can easily access pertinent information. This pilot focuses on banks regulated by the Monetary Authority of Singapore (MAS), or their affiliates. Through registration of security tokens, the system provides a streamlined and unified solution for investors and market participants to access information about the digital assets, including the platforms they may be trading on.

9.3.2 Objectives

The key objectives of the Digital Bond Listing framework and the Digital Assets Register pilot include:

- Enabling the listing of digital bonds and promoting the integration of such bonds into mainstream capital markets.
- Extending the wholesale bond listing framework to include digital bonds, which will be subject to the regulatory requirements for listing and ongoing disclosures.
- Establishing the Digital Assets Register as a centralised repository to maintain and showcase security token issuance data.
- Creating greater transparency in the digital assets landscape by allowing investors to easily access information about security tokens.
- Building an easily accessible platform to facilitate the growth of digital securities issuance.

9.3.3 Solution design

Digital Bond Listing

Digital Bond Listing utilises the existing SGX listing framework for wholesale debt securities. Similar to the listing process for conventional bonds, listing applications for digital bonds can be submitted by a listing agent through the Singapore Exchange Regulation (SGX RegCo) online submission system.

Benefits of listing a digital bond include:

- For issuers, enhanced visibility and distribution reach to investors, as well as higher credibility among investors through greater transparency.
- For investors, they will be able to take comfort from the regulatory requirements for listing and ongoing disclosures, as well as easily access relevant information about listed digital bonds.

Issuers of listed bonds will be expected to set out the list of risk factors that investors should be aware of in the offering document, such as business, market, geopolitical, liquidity and legal risks. For digital

bonds, disclosures about the tokenisation platform and technological risks are also expected to be included in the offering document.

More details can be found on <https://www.sgx.com/fixed-income/digital-bond>.

Digital Assets Register

The Digital Assets Register serves as a centralised repository of information easily accessible via the SGX Fixed Income webpage. The implementation approach is via online submission for registration on a dedicated page: <https://www.sgx.com/fixed-income/digital-assets-register>.

The Digital Assets Register will include important information such as issuer name, security identifier and the offering size. There will also be offer documents related to the issuance, which would contain information including the technology stack, risk factors and identities of the involved parties.

9.3.4 Key Achievements, learnings and future developments

Digital Bond Listing

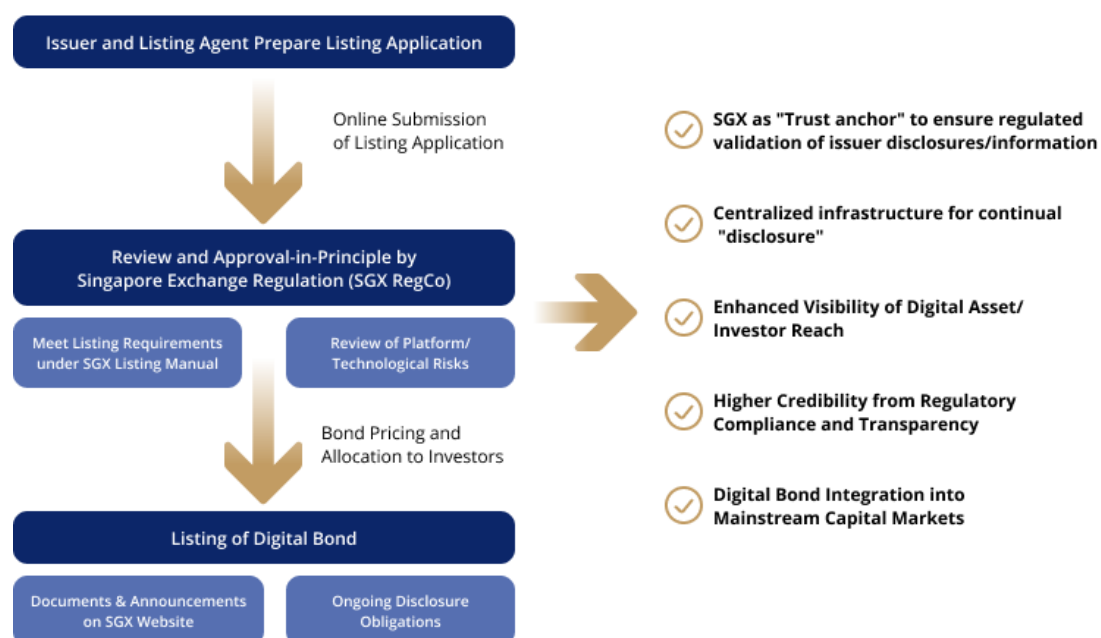


Figure 7: Overview of SGX Digital Bond Listing Framework

- The extension of SGX's wholesale bond listing framework to include digital bonds is a step towards promoting regulatory compliance and security in this rapidly evolving market.
- Issuers will be able to enjoy the benefits of listing their digital bonds, similar to traditional bonds, including enhanced visibility and profile and access to a larger pool of investors.
- Successful consultation with industry stakeholders on regulatory aspects.

Digital Assets Register

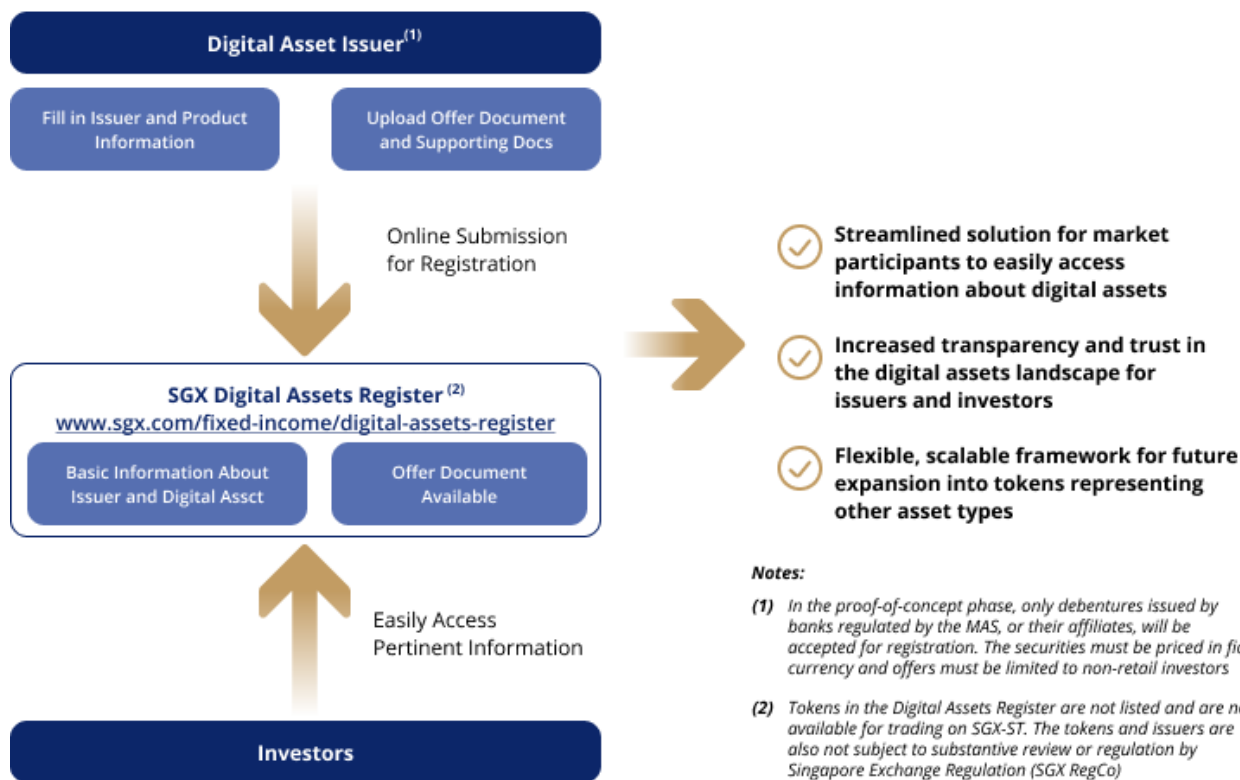


Figure 8: Overview of SGX Digital Assets Register

- Streamlined process for accessing and managing security token information, enhancing transparency and trust for issuers and investors.
- A flexible, scalable framework for future expansion into security tokens representing other asset types, further promoting blockchain technology within financial markets.

Future developments

- Through Project Guardian, SGX seeks to collaborate with other financial industry stakeholders to promote innovation in fixed income markets. As a market operator and infrastructure provider, SGX is keen to work together with other stakeholders to integrate tokenised fixed income securities into mainstream capital markets, and to encourage interoperability between traditional and tokenised fixed income securities.
- For the Digital Assets Register, expansion may include additional types of digital assets beyond debentures, as part of efforts to diversify and deepen the digital asset markets.
- SGX may also consider collaborating with international regulatory bodies or financial institutions to explore how the Digital Assets Register could be a global solution as a repository for information about digital assets.

10 Conclusion

The adoption of asset tokenisation in fixed income markets, as evidenced by pilots conducted under Project Guardian has highlighted the transformative potential of tokenised financial instruments in DCM. Tokenisation offers substantial operational benefits, including reduced counterparty risks, automated repo transactions (e.g. intraday repo / liquidity instruments), streamlined workflow for bond issuance and asset lifecycle management processes. By enabling more efficient cross-border transactions, tokenisation can foster greater financial integration and collaboration among international markets, ultimately contributing to global economic growth.

This report discussed the potential for tokenised fixed income securities to increase efficiency and transparency of existing fixed income markets. It also highlighted key risk considerations and mitigating actions needed for tokenised fixed income security-based issuances and lifecycle management. A unified approach on these risk mitigations is needed to facilitate acceptance across a large set of participants. Tokenised debt issuances will be able to scale across global markets only when established and accepted best practices are recognised across all market participants.

To fully capitalise the potential of tokenisation in fixed income markets and beyond, the report makes the case for the establishment and acceptance of industry standards. In this report, ICMA's BDT and the CMTAT were introduced as emerging industry standards that have been adopted by financial institutions. The ongoing development of industry standards is instrumental in ensuring interoperability across tokenised and traditional markets, facilitating seamless integrations along trade life cycle.

The report introduced GFIF as a framework for determining potential operating and data model of tokenised fixed income products. The framework, as a principle, advocates building on existing standards and extending it where new formats and requirements are needed. The application of the technology and jurisdiction-agnostic design principles as outlined in this report, seeks to provide a robust and comprehensive framework ensuring the legal validity, operational integrity, and transparency of tokenised securities. This facilitates a more predictable and secure environment for investors and issuers, promoting trust and wider acceptance of tokenised financial products. In addition, the application of the design principles will be a key enabler of interoperability across different product frameworks and operating models.

The acceptance of tokenised fixed income securities and smart contracts is poised to grow and increase efficiency of global financial ecosystems. Future developments will focus on enhancing interoperability between traditional and digital financial infrastructures. The ongoing development of regulatory and technical standards will be crucial in this process, ensuring that the ecosystem remains robust and adaptable.

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12 Appendix

12.1 ICMA's Bond Data Taxonomy Overview

12.1.1 Asset information

Data elements to define a debt security in the BDT include the following data fields and attributes.

Security identifier and status	Example input (non-exhaustive)	Field type	Data type
Identifier Type	ISIN, SEDOL, WKN, DTI, CMU...	list	string
Identifier Value	XS2524143554	predefined	string
Issue Ratings Agency	Fitch, Moody's, S&P...	list	string
Expected Issue Ratings Value	A+	free	string
Form of the Note	Registered, Bearer, Fed Bookentry...	list	string
Status of the Notes	Senior Secured, Senior Non Preferred...	list	string
DLT Bond Indicator	TRUE	predefined	Boolean

Issuer	Example input (non-exhaustive)	Field type	Data type
Issuer Name	Coöperatieve Rabobank U.A.	free	string
Issuer LEI	DG3RU1DBUFHT4ZF9WN62	predefined	string
Issuer Bloomberg Ticker	OBL	free	string
Issuer Ratings Agency	Fitch, Moody's, S&P...	list	string
Issuer Ratings Value	A+	free	string
Issuer Ratings Outlook	Positive, Stable, Negative....	list	string

Guarantor [if applicable]	Example input (non-exhaustive)	Field type	Data type
Guarantor Name	UBS	free	string
Guarantor LEI	7245009UXRIGIRYOBR48	predefined	string
Guarantor Ratings Agency	Fitch, Moody's, S&P...	list	string
Guarantor Ratings Value	A+	free	string
Guarantor Ratings Outlook	Positive, Stable, Negative....	list	string

Amounts	Example input (non-exhaustive)	Field type	Data type
Specified Denomination	100,000.00	predefined	decimal
Integral Multiples	100,000	predefined	decimal
Calculation Amount	1,000	predefined	decimal
Final Redemption Amount	100%	predefined	decimal
Early Redemption Amount	100%	predefined	decimal
Aggregate Nominal Amount	1,500,000,000.00	predefined	decimal
Outstanding Amount Before Increase	750,000,000.00	predefined	decimal
Outstanding Amount After Increase	750,000,000.00	predefined	decimal

Series Number	472	free	string
Series Amount	750,000,000.00	predefined	decimal
Tranche Number	1	free	string
Tranche Amount	750,000,000.00	predefined	decimal
Net Proceeds	750,000,000.00	predefined	decimal
Gross Proceeds	760,000,000.00	predefined	decimal

Currency	Example input (non-exhaustive)	Field type	Data type
Specified Currency	EUR	list	string
Payment Currency	EUR	list	string
Settlement Currency	EUR	list	string

Interest & Calculations	Example input (non-exhaustive)	Field type	Data type
Interest Type	Fixed, Floating, Zero coupon...	list	string
Interest Payment Frequency	Annually, Semi-annually...	list	string
Interest Rate	3.875	predefined	decimal
Accrued Interest	10000	predefined	decimal
Interest Rate Benchmark	SOFR, EONA,	list	string
Term	3	predefined	integer
Term Period	MNTH	list	string
Calculation Type	Compounded Daily, Weighted Average	list	string
Determination Method	Calculated	list	string
Observation Method	Observation Shift, Lock-out...	list	string
Period	3 MNTH	list	string
Margin (%)	100	predefined	decimal
Day Count Fraction	ICMA Act/Act	list	string
Business Days	TARGET2	list	string
Business Day Convention	Following, Unadjusted	list	string
Interest Commencement Date	2021-02-22	calendar	dateformat
First Interest Payment Date	2022-02-22	calendar	dateformat
BrokenDateType	Short	list	string
Last Interest Payment Date	2026-02-22	calendar	dateformat
PayableDate	D=15, M=6	predefined	integer
Maturity Date	2031-02-22	calendar	dateformat
Change of Interest/Payment Basis	For the period from (and including) the Interest Commencement Date, up to (but excluding) 2021-08-15 paragraph 14 applies.	free	string

12.1.2 Transaction and settlement data

Parties	Example input (non-exhaustive)	Field type	Data type
Lead Manager	LEI-ANGGYXNX0JLX3X63JN86	ID-Link	string
Joint Lead Managers	LEI-7245009UXRIGIRYOB48	ID-Link	string

Joint Global Coordinators	LEI-8IBZUGJ7JPLH368JE346	ID-Link	string
Joint Bookrunners	LEI-BFM8T61CT2L1QCEMIK50	ID-Link	string
Stabilisation Manager	LEI-DG3RU1DBUFHT4ZF9WN62	ID-Link	string
Fiscal Agent	LEI-DG3RU1DBUFHT4ZF9WN62	ID-Link	string
Principal Paying Agent	LEI-DG3RU1DBUFHT4ZF9WN62	ID-Link	string
Calculation Agent	LEI-DG3RU1DBUFHT4ZF9WN62	ID-Link	string
Trustee	LEI-DG3RU1DBUFHT4ZF9WN62	ID-Link	string
Clearing/Settlement System	Euroclear Bank SA/NV	list	string
Other Party Roles			

Dates	Example input (non-exhaustive)	Field type	Data type
Issue Date	2021-02-15	calendar	dateformat
Pricing Date	2021-02-15	calendar	dateformat
Settlement Date	2021-02-22	calendar	dateformat

Pricing Data	Example input (non-exhaustive)	Field type	Data type
Issue Price	99.841	predefined	decimal
Indication of Yield	3.912	predefined	decimal
Fees	1.1	predefined	decimal
All-in Price	99.841	predefined	decimal
All-in Yield p.a.	3.912	predefined	decimal
All-in Spread vs. Benchmark	296	predefined	decimal
Reoffer Price	99.841	predefined	decimal
Reoffer Yield p.a.	3.912	predefined	decimal
Benchmark Bond Ticker	OBL	predefined	string
Benchmark Bond Interest Rate	1.3	predefined	decimal
Benchmark Bond Maturity Date	2027-10-15	calendar	dateformat
Benchmark Bond ISIN	DE0001141869	predefined	string
Benchmark Bond Price	100.97	predefined	decimal
Benchmark Bond Yield p.a.	1.104	predefined	decimal
Reoffer Spread vs. Benchmark	296	predefined	decimal
Mid Swap Rate	1.1952	predefined	decimal
Reoffer Spread vs. Mid-Swaps	104.8	predefined	decimal
Expenses	Issuer pays legal fees	free	string
Fee Allocation Type	Equal, Proportional	list	string

Optionality	Example input (non-exhaustive)	Field type	Data type
Call/Put Option	Not Applicable	list	string
Optional Redemption Dates	2022-02-22	calendar	dateformat
StartDate	2027-08-30	calendar	dateformat
EndDate	First Reset Date	predefined	string

Redemption Payment Basis	Partly Paid	list	string
Optional Redemption Amount	750,000,000.00	predefined	decimal
Yield to Call	3.912	predefined	decimal

12.1.3 Compliance and regulatory data

Listing and Restrictions	Example input (non-exhaustive)	Field type	Data type
Listing	Copenhagen Stock Exchange	list	string
Listing Market Type	RMKT	list	string
Method of distribution	Syndicated	list	string
Governing Law	Luxembourg Law	list	string
Documentation Name	Issuer's Debt Issuance Program	free	string
Documentation Date	2021-02-22	calendar	dateformat
Documentation Link	https://document	free	string
Selling Restrictions	Reg. S Compliance Category 2	list	string
PRIPs Restriction	Prohibition of sales to UK retail investors	list	string
Additional Selling Restrictions	UK, EEA and US and as further set out in the Offering Memorandum	free	string
Manufacturer Target Market	UK MIFIR product governance / Professional investors and ECPs only target market	list	string
Negative Pledge	Not Applicable	list	string
Gross Up	No gross up if tax deduction or withholding is imposed	free	string
Cross Default	Not Applicable	list	string
Substitution or variation	Not Applicable	list	string
Use of Proceeds	For Eligible Projects as defined under the Green Bond Framework, as further described in the Offering Memorandum.	free	string

12.1.4 DLT-related information

The DLT includes a “DLT extension” which enables users to capture DLT-related information of a digital (DLT-based) bonds in a structured format. While many data points are not standardised and vary by use case and jurisdiction, the BDT provides umbrella terms and data specifications that facilitate the exchange of information.

"DLT extension", referring to DLTPlatformInformation Block	Example input (non-exhaustive)	Field type	Data type
Type	Ethereum, Hyperledger Besu, Hyperledger Fabric, R3 Corda, Canton...	free	string
Accessibility	Public, Private...	list	string

Role	The Notes being registered in a DLT constituting a dispositif d'enregistrement électronique partagé, as defined in Articles L. 211-3 et seq. of the French Monetary and Financial Code (the DEEP).	Free (from prospectus)	string
Fee	One Time GBP 10,000	free	string
Platform Operator	SG Forge, HSBC Orion, GSBE, Euroclear...	ID-Link	string
Name	DEEP	free	string
Identifier	MIC=SGF, DTI=XC7KDF8R9	predefined	string
Token Type	Security, CBDC....	free	string
Token Technical Reference	ERC-20, ERC-2020	free	string
Token Technical Reference ID	BD7KDF8R3, XS2524143554	list	string
Smart Contract	0xE9Ac9Eb7cDDAafd9e5CA43c23bd521aFcDK8BE49	free	string

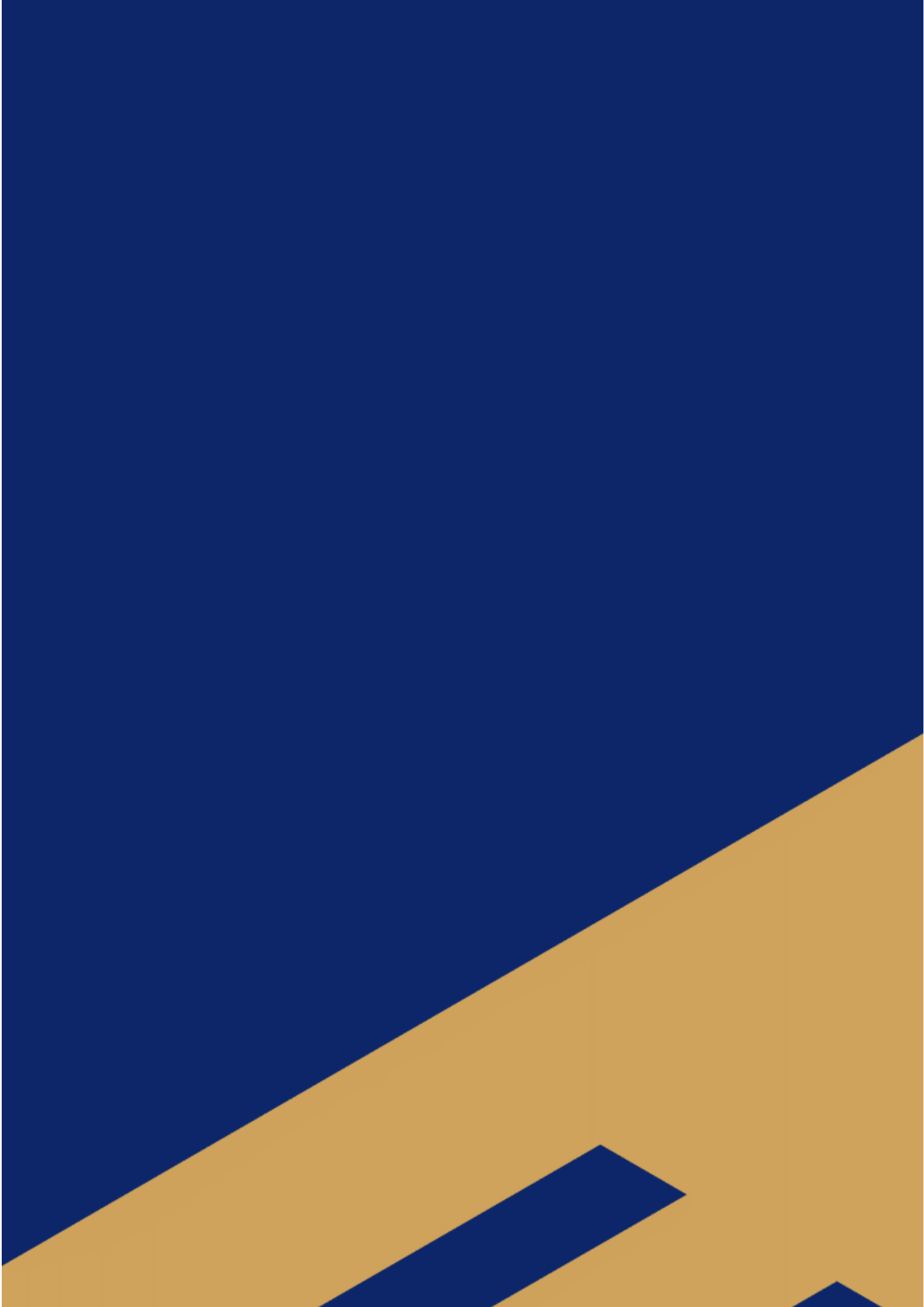
12.1.5 Contractual terms

The offering document, also referred to as a prospectus or offering memorandum, sets out the terms and conditions of a digital (DLT-based) debt security offering. From a data model perspective, the names and related attributes of transaction parties, numerical values and interest calculation methodologies are represented in the BDT. Other restrictions such as eligible investors and conditions such as negative pledge or tax-related aspects can be captured and communicated in a simplified, binary form, to provide an indication as to whether such conditions exist. The details of such conditions are described in the relevant documentation, which can be referenced in the group of fields “documentation name”, “documentation date”, and “documentation link”.

However, other contractual terms specific to digital (DLT-based) bond transactions may include:

- Bilateral agreements with DLT services providers, for example to govern access to the DLT platform or network, define the roles and responsibilities of each party, and ensure compliance with relevant local laws and regulations.
- Bespoke custody agreements for native digital or tokenised securities under applicable laws.
- The GMRA or Global Master Securities Lending Agreement (GMSLA), which include a Digital Assets Annex, for parties in a cross-border repo or securities lending transaction.

Contractual terms, whether based on a digital (DLT-based) bond offering document or bilateral agreements, form the basis for automating workflows or translating legal obligations and requirements into smart contracts i.e. computer code. From a technical perspective, the BDT’s data definitions and specifications can be used for different smart contracts and workflows to define a digital (DLT-based) bond consistently and facilitate the exchange of data on-chain and off-chain across different platforms and architectures.



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The background of the page features a large, abstract geometric design. It consists of several overlapping triangles in two colors: a deep navy blue and a mustard yellow. The design is minimalist and modern, with sharp lines and a clean aesthetic. The blue triangles are positioned in the upper left and lower right areas, while the yellow triangles fill the remaining space, creating a dynamic and balanced composition.